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# COMPUTER AIDED DIAGNOSIS OF PANCREATIC CANCER USING CT IMAGES

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# Summary

Recent years have seen considerable progress in medical image analysis, primarily propelled by advancements in machine learning and tools for computer-aided diagnosis. Among multiple health issues, pancreatic disorders, especially pancreatic cancer, present significant obstacles for early detection owing to their subtle and intricate signs in imaging data. Timely and precise diagnosis is crucial for enhancing patient outcomes, and automated image classification techniques present a hopeful approach to assist clinicians in their decision-making process. This thesis explores the use of machine learning methods to classify pancreatic and non-pancreatic medical images, focusing on methodological rigor and practical performance assessment. The dataset utilized in this research includes medical images from pancreatic and non-pancreatic areas. For thorough model assessment, the dataset is split into distinct training and testing portions. The preprocessing pipeline is meticulously crafted to ready the images for efficient feature extraction and classification. Essential preprocessing tasks involve transforming the images into one-dimensional numerical vectors and forming structured data matrices appropriate for feeding into machine learning algorithms. Moreover, a group of twenty images is randomly chosen from the test set to aid in exploratory data analysis, which includes calculating correlation matrices and assessing pixel intensity distributions. This examination seeks to recognize patterns and distinctions between pancreatic and non-pancreatic images, offering insights into characteristics that could be significant for classification. A thorough analysis of pixel intensity values is performed to measure the differences between pancreatic and normal images. The average image brightness distribution is examined to evaluate its capability as a distinguishing attribute. Structural similarity (SSIM) metrics are calculated to assess the level of visual resemblance between images, providing an extra quantitative viewpoint on image attributes. These evaluations together establish the foundation for a thorough comprehension of the image dataset and guide the choice of features for subsequent classification tasks. The classification framework in this research combines conventional and sophisticated machine learning models. Support Vector Machines (SVM) are utilized for their efficiency in dealing with high-dimensional data and their capacity to identify optimal decision boundaries. XGBoost, a robust ensemble learning method recognized for its predictive capabilities, is also utilized to assess performance among various classifier frameworks. The processed training set is used to train both models, and their performance is assessed on the testing set through standard metrics like accuracy, precision, recall,

and F1-score. This thorough assessment allows for an essential comparison of classifier performance and offers insights into the most effective methods for classifying pancreatic images. This study offers multiple contributions to the area of medical image analysis. Initially, it offers a structured preprocessing and feature extraction approach that converts unrefined imaging data into organized numerical formats appropriate for machine learning uses. Additionally, it shows the usefulness of exploratory analyses—such as pixel intensity distribution, average brightness, correlation matrices, and SSIM metrics—for uncovering distinguishing features and comprehending dataset attributes. Third, utilizing both SVM and XGBoost classifiers provides a comparative viewpoint on various machine learning paradigms, emphasizing the advantages and drawbacks of each in the realm of pancreatic image classification. The results of this thesis highlight the possibilities of merging statistical image analysis with strong machine learning methods to create automated, dependable, and understandable diagnostic instruments. Although concentrated on pancreatic imaging, the methods and findings discussed here hold wider significance for medical image analysis and the creation of computer-assisted diagnostic systems in multiple clinical areas. Utilizing the advantages of data-driven feature analysis along with sophisticated classification models, this study adds to the expanding research focused on enhancing the precision, effectiveness, and accessibility of medical diagnostics.

# Contents

# Chapter 1

# Introduction

Pancreatic cancer is widely recognized as one of the most lethal cancers globally. The overall five-year relative survival rate for pancreatic cancer, as reported by the American Cancer Society's "Cancer Statistics, 2025," is approximately 13% with an increase of 1% per year. Pancreatic cancer is the fourth leading cause of cancer in the world with more than 200 000 deaths every year. Increase in pancreatic cancer mortality has also been reported in European populations as one of the highest. Most pancreatic cancers (over 90%) are exocrine tumors, which arise from the ducts that carry digestive enzymes (ductal adenocarcinoma).Pancreatic cancer has a low survival rate due to late diagnosis. The overall 5-year survival rate is 5-15%. For early-stage tumors that are surgically removed, the rate rises to about 20%. Neuroendocrine tumors have a better prognosis, with a 5-year survival rate of 65%. This low survival rate is largely attributed to the fact that the disease often presents with non-specific symptoms in its late stages, making early detection a significant challenge. Understanding risk factors is crucial for identifying individuals who may benefit from enhanced surveillance. High fasting plasma glucose remains the major risk factor influencing the incidence, mortality, and prevalence of pancreatic cancer, followed by tobacco use and high body mass index. People with obesity (BMI of 30 or more) are about 20% more likely to develop pancreatic cancer , with weight distribution around the waistline being particularly concerning. Tobacco use is the main risk factor for pancreatic cancer, with pancreatic adenocarcinoma being 2 to 3 times more common in heavy smokers than in nonsmokers. The risk increases with both the amount and duration of smoking, though cessation can gradually reduce this elevated risk over time. Genetic factors, such as BRCA1, BRCA2, PALB2, and KRAS mutations, alongside lifestyle, environmental, and medical conditions, play pivotal roles in pancreatic cancer development Risk factors that have been scientifically proven to have biological or genetic links to pancreatic cancer include cigarette smoking, chronic pancreatitis, and family history. Individuals with hereditary cancer syndromes, such as Lynch syndrome or familial pancreatic cancer, face substantially elevated risks and may benefit from screening protocols. Additional modifiable risk factors include chronic pancreatitis, diabetes mellitus (both as a risk factor and potentially an early manifestation of the disease), heavy alcohol consumption, and occupational exposures to certain pesticides and chemicals. Age is also a significant non-modifiable risk factor, with most cases diagnosed in individuals over 65 years old. Pancreatic cancer is characterized by the uncontrolled growth of cells in the pancreas, an organ crucial for both digestion and blood sugar regulation. This uncontrolled growth can form a pancreatic tumor. Early detection of a small, localized tumor is critically important because surgical resection is the only chance for a curative outcome. The detection of these small, resectable tumors can drastically reduce the number of cancers. The most common type of pancreatic cancer, pancreatic adenocarcinoma (PDAC), accounts for over 95% of all pancreatic tumors. PDAC most frequently develops in the head of the pancreas, which can lead to earlier symptom onset due to bile duct obstruction, though these symptoms are often mistaken for other conditions. The aggressive nature of PDAC is attributed to several factors, including its tendency to invade local blood vessels and nerves early in disease progression, its dense desmoplastic stroma that limits drug penetration, and its propensity for early micrometastatic spread. The tumor microenvironment in pancreatic cancer is uniquely immunosuppressive, contributing to poor treatment responses and rapid progression. Pancreatic cancer is often called a "silent disease" because symptoms typically appear only after the disease has advanced or spread. While analyzing a patient with pancreatic adenocarcinoma, there are several key factors which need to be taken care of. Like other types of tumors, these are also staged at several levels. However, detecting these cancers at an early stage is nearly impossible. An open case study on pancreatic cancer suggests that there are 12 alarm systems: weight loss, abdominal pain, nausea and vomiting, bloating, dyspepsia, new-onset diabetes, changes in bowel habit, pruritus, lethargy, back pain, shoulder pain, and jaundice. Out of these, five symptoms have been shown to occur more than 6 months before diagnosis: back pain, shoulder pain, dysphagia, changes in bowel habit, and lethargy. Lethargy is a medical term used for extreme tiredness or lack of energy which can be described as depression. The location of the tumor significantly influences symptom presentation. Tumors in the pancreatic head often cause jaundice earlier due to bile duct obstruction, while body and tail tumors may remain asymptomatic longer, typically presenting with more advanced disease. New-onset diabetes in adults over 50, particularly when accompanied by weight loss, warrants investigation for underlying pancreatic malignancy, as diabetes can be both a risk factor and an early manifestation of pancreatic cancer. Several advanced imaging tests are crucial for diagnosis and staging. While transabdominal ultrasound is a common screening tool, it is usually limited in the evaluation of the pancreas due to the presence of adipose tissue and intestinal and gastric bloating. More definitive methods include CT scans, MRI, and PET scans, which provide detailed images of the pancreas and help determine the extent of the disease. However, endoscopic ultrasound (EUS) is the first non-invasive imaging test for evaluation of the pancreas which provides high-resolution images and is a commonly accepted sensitive technique for detection of small pancreatic head tumors. EUS is particularly valuable for detecting small tumors and facilitating biopsies through fine-needle aspiration (FNA), allowing for cytological or histological confirmation. The combination of EUS with FNA has significantly improved diagnostic accuracy, particularly for lesions less than 2 cm in diameter. Magnetic resonance cholangiopancreatography (MRCP) is another valuable non-invasive technique that can visualize the pancreatic and bile ducts without the need for contrast injection, helping to identify ductal abnormalities and assess resectability. Endoscopic retrograde cholangiopancreatography (ERCP) serves both diagnostic and therapeutic purposes, allowing for tissue sampling and biliary stent placement in cases of obstruction.

## 1.1 Computed Tomography

Computed tomography refers to an x-ray imaging procedure in which a narrow beam of x ray is aimed at a patient and rotated around the body producing signals that are processed by the machine and later generate cross-section images. These cross-section / slices are called tomographic images and give us detailed information whether the patient has been affected by cancer or not. In terms of pancreatic cancer, During the computed tomography procedure the patient lays down on a bed that slowly moves through the gantry while the x ray tube scans through the body. After every rotation the CT computer uses mathematical techniques to determine the image slice of the patient, the thickness of the tissue and the location. Each slice of the of the patient ranges from 1-10 millimetres. This process is carried out until desired number of slices are collected. Image slices can either be displayed individually or stacked together by the computer to provide a 3D image of the affected location. Key advantages of computed tomography in pancreatic cancer is its ability to detect spread to nearby organs, lymph nodes and evaluate vascular involvement. Compared to MRI, CT has shown better performance for vascular staging in pancreatic tumors by using 3D isotropic, minimum intensity projection and maximum intensity projection. Despite the increasing knowledge on human body, pancreatic cancer is still undiagnosable due to late recognition. Computed tomography plays a vital role in this due to its advancement which gives us an opportunity to perform surgery, staging and chemotherapy before time. The primary motivation of this thesis is to contribute to the ongoing research on pancreatic tumor detection by enhancing the diagnostic potential of CT imaging. Specifically, this work focuses on evaluating and analysing CT data through advanced computational approaches. To achieve this, I employed methodologies such as visualization graphs, Support Vector Machines (SVM), Boosting algorithms, and the deep learning model AlexNet. By integrating these methods, this thesis aims to strengthen early detection strategies and provide insights into the role of CT imaging in pancreatic cancer diagnosis.

# Chapter 2

# Related Work

Several previous studies have explored computational approaches for pancreatic tumor detection. [AA25]In a study by Yusuf Alaca and Omer Faruk Akmese, the authors developed a novel framework based on Generative Adversarial Networks (GANs). This model was designed to distinguish between real and synthetically generated images. The study began with the use of the Harris Detection Algorithm to identify important corner points. DenseNet121, a deep convolutional neural network with 121 layers, was then employed to combine two images and extract the most informative features. The authors also integrated InceptionV3 to further enhance performance. For classification, K-Nearest Neighbours (K-NN), Support Vector Machines (SVM), and Random Forest (RF) were tested. [Nad+25] Another significant contribution came from Abubakar Nadeem, Rahan Ashraf, Toqeer Mahmood, and Sajida Parveen, who worked on an automated computer-aided diagnosis system for pancreatic tumour detection using CT scans. Their framework consisted of four stages: preprocessing, segmentation, detection, and classification, implemented with an 11-layer AlexNet convolutional neural network. Data augmentation was applied by flipping images between –30° and 45° to increase robustness and reduce overfitting. Images were resized to 227 × 227 pixels and converted to grayscale based on pixel intensity. Preprocessing also included Sobel edge detection to capture intensity gradients and watershed segmentation to separate tumor regions by applying region-based shading. These steps were followed by morphological operations, such as dilation (to modify shape and structure) and erosion (to shrink object boundaries and remove noise). Finally, feature extraction was carried out using the Gray Level Co-occurrence Matrix (GLCM) for texture analysis, which captured discriminative patterns for tumor classification. [Cao+23] A more advanced framework was introduced in another study, which proposed a three-stage deep learning pipeline for pancreatic cancer detection using non-contrast CT (NCCT) scans. The first stage involved pancreas localization using a U-Net segmentation model to isolate the pancreas from surrounding anatomical structures, reducing noise and computational complexity. In the second stage, a lesion detection network was applied within the localized pancreas region, with very high specificity (~99%) to minimize false positives. The third stage used a differential diagnosis network enhanced by a memory-augmented transformer module, which stored representative feature prototypes from prior cases. This addition improved the model's ability to distinguish pancreatic ductal adenocarcinoma (PDAC) from seven other non-PDAC lesions. The system was trained on an internal dataset of over 3,000 patients, where lesion annotations were created by registering contrast-enhanced CT scans. Validation was performed on external datasets from nine medical centres (over 5,000 patients), and retrospective deployment was conducted on more than 20,000 NCCT scans from routine clinical workflows. This ensured high robustness, scalability, and clinical applicability. In addition to imaging-based approaches, several studies have highlighted risk factors and alternative screening methods related to pancreatic cancer. Pancreatitis, often caused by alcohol consumption or gallstones, has been identified as a condition that increases the risk of pancreatic cancer. Among screening methods, the N-NOSE plus test has shown promising performance, with an AUC of 0.856. Furthermore, epidemiological studies have revealed that individuals with blood group A are at higher risk due to increased production of the A antigen (especially the A1 allele). Conversely, blood group O is associated with the lowest risk, as the absence of antigen production reduces enzyme activity linked to tumor development. Diabetes has also been strongly correlated with pancreatic cancer risk, with up to 80% of cases being detected in the pre-symptomatic phase. Imaging techniques such as MRI, CT, and ultrasound remain essential diagnostic tools in this context. Across these studies, several limitations have been identified, including reliance on relatively small and less diverse datasets, challenges in early-stage tumor detection, and generalizability across patient populations. These factors highlight the need for further research to enhance CT-based diagnostic approaches and integrate advanced computational techniques for more reliable pancreatic tumor detection. [Vai+22]The Intelligent Deep-Learning-Enabled Decision-Making Medical System for Pancreatic Tumor Classification (IDLDMS-PTC), as conceived by Vaiyapuri et al., represents a robust integration of classical image processing and advanced deep learning techniques, specifically engineered to maximize the diagnostic performance on challenging CT images. The workflow begins with Gabor Filtering (GF), which serves a critical pre-processing role by leveraging the filter's dual localization property—both in the spatial and frequency domains—to perform highly effective texture analysis while simultaneously mitigating image noise. Following this, the challenging task of accurately isolating the tumor is handled by the Emperor Penguin Optimizer (EPO) guiding a Multilevel Thresholding (MLT) technique, forming the EPO-MLT segmentation block; this innovative optimization-driven segmentation is crucial for tackling the often subtle and indistinct boundaries of pancreatic lesions, classifying the gray level image into k+1 distinct classes. The segmented output then transitions to the core feature extraction phase, where the MobileNet architecture is employed not just for its deep learning capabilities, but specifically for its computational efficiency and speed. This efficiency stems directly from its use of depth-wise separable convolutions, which factor a standard 3D convolution into two distinct, more resource-friendly steps: a channel-wise depth-wise convolution at the filter phase, followed by a 1×1 point-wise convolution phase to combine the outputs, thereby accelerating the generation of feature vectors compared to traditional convolutional networks. Finally, the extracted, high-quality features are fed into an Autoencoder (AE) for the final detection and classification decision. The pinnacle of the system's optimization lies in the use of the Multileader Optimization (MLO) algorithm, a meta-heuristic approach employed to precisely tune the sensitive internal weight and bias parameters of the AE classifier, ensuring the model's predictive capability is maximally leveraged. This comprehensive and synergistically optimized design resulted in the IDLDMS-PTC achieving an exceptional average performance metric of 0.9935, confirming its effectiveness and positioning it as a powerful tool for computer-aided diagnosis in pancreatic oncology. [Bab+24] Babaei and their team in 2024 came up with a new way to find tumors in the pancreas. They treated the problem like finding unusual parts in images. They trained their system using the MSD pancreas dataset and tested it on another dataset from the University of Oklahoma Health Sciences Center. Before training, they did some steps to prepare the images, like finding where the pancreas is, cutting out the relevant parts, and making the images clearer. Then they used a special type of model called a denoising diffusion model to create clean versions of the scans. They also added a simple tool to help the model better find the tumors by comparing the original scans with the cleaned-up ones. This helped them create maps that showed where the tumors were, which they then turned into final images showing the tumor areas. Even though their method didn't perform as well as some other models that use more detailed information, it still worked well in real clinical cases. The main advantage is that it only needs simple labels for the images, which is easier to get than detailed markings. However, the results depended a lot on the settings they used, and the model didn't always get the edges of the tumors exactly right. [Muk+25]Mukherjee et al. proposed a comprehensive deep learning framework designed to perform automated pancreas segmentation from CT scans, with the broader aim of supporting early pancreatic cancer detection and biomarker discovery. For this study, the authors collected the largest single-institution CT dataset available to date, consisting of 3,031 portal venous phase scans. Each scan was manually annotated by radiologists using a standardized protocol, which ensured reliable and consistent ground truth segmentations for training. The model was built on a 3D U-Net architecture, a self-configuring U-Net framework that automatically adapts its parameters to the dataset. To improve robustness, the authors employed extensive augmentation strategies such as geometric transformations, brightness and contrast adjustments, and noise injections, thereby simulating real-world imaging variations. Training was conducted using five-fold cross-validation, and the final framework was constructed as an ensemble of these folds to maximize predictive accuracy. For evaluation, the framework was tested not only on the internal dataset but also on multi-institutional external datasets. On the internal test set, the model achieved a Dice similarity score of approximately 0.94, while on the AbdomenCT-1K dataset, which consisted of 585 CT scans from multiple centers, the performance was even higher, with a Dice score of ˜0.96.

These results demonstrated the robustness of the model across diverse scanners and imaging protocols. When evaluated on the NIH-PCT dataset, the performance decreased slightly to ˜0.87, though the authors observed that in several challenging cases the model's segmentations were qualitatively superior to the existing annotations, suggesting the potential limitations of the ground truth itself. Beyond quantitative metrics, the study highlighted the clinical relevance of accurate pancreas segmentation, as it enables precise volumetric analysis and the extraction of imaging biomarkers, both of which are critical for detecting subtle morphological changes that may signal early stages of pancreatic cancer. The authors concluded that their framework, by reliably segmenting the pancreas at scale, could serve as a powerful tool for large-scale research into pancreatic disease and early cancer detection. They also noted, however, that the model was trained primarily on scans of morphologically normal pancreases and may require further refinement when applied to cases with diseased or atypical morphology, such as chronic pancreatitis or advanced tumors. Despite these challenges, the study demonstrated that automated pancreas segmentation with deep learning can not only achieve state-of-the-art accuracy but also contribute meaningfully to clinical workflows and large-scale studies of pancreatic health.[CD25] Cereda and D'Andrea (2025), in their review "Pancreatic Cancer: Failures and Hopes – A Review of New Promising Treatment Approaches", analyse the persistent challenges and emerging strategies in the management of pancreatic cancer. The authors emphasize that despite significant advancements in systemic therapies, pancreatic cancer remains one of the deadliest cancers, with a five-year survival rate of only 3% in advanced stages. This poor prognosis is attributed to multiple factors, including late-stage diagnosis due to the lack of early symptoms, high genetic heterogeneity—especially KRAS mutations that drive tumor progression and therapy resistance—and the presence of a dense, fibrotic, and immunosuppressive tumor microenvironment that hinders drug delivery and limits the effectiveness of immunotherapy. To address these challenges, the review highlights several innovative approaches under investigation. Sequential therapy cycles are being explored to overcome resistance mechanisms by alternating or combining different systemic treatments. The identification of predictive biomarkers is emphasized to personalize therapy, targeting specific patient subpopulations likely to respond to interventions. Combination therapies are also discussed, which aim to simultaneously target tumor cells and modulate the tumor microenvironment to enhance treatment efficacy. Additionally, the development of advanced drug delivery systems is being pursued to effectively penetrate the desmoplastic stroma and improve local drug concentrations at the tumor site. The authors conclude that while pancreatic cancer treatment remains highly challenging, these emerging strategies provide a promising direction for improving therapeutic outcomes, survival, and quality of life for patients. Review in Biomedicines highlights the critical need for early detection of pancreatic cancer. Most people are diagnosed when the disease is already advanced, making it hard to treat and leading to very low survivalrates. The article points out that one big problem in catching the disease early is that it does't show symptoms in the early stages. This makes it difficult to detect using standard imaging tests or clinical checks. To tackle this, the review looks at new methods based on biomarkers. These include circulating tumor DNA, microRNAs, and exosomes, which are all promising signs that can be found in blood or other body fluids. These biomarkers can help spot changes linked to early tumor growth without the need for invasive procedures like tissue biopsies. The review also talks about focusing on high-risk groups for early screening. These groups include people over 50 who recently developed diabetes, those with chronic pancreatitis,individuals with pancreatic fat buildup, people with cysts in the pancreas, and those with inherited cancer conditions. Paying more attention to these groups can greatly improve the accuracy of early detection. The article also suggests combining biomarker tests with imaging techniques such as MRI, CT scans, and endoscopic ultrasound. This mix of molecular and physical imaging might lead to better diagnostic results. Overall, the review says that using new biomarkers, advanced imaging, and targeted screening could help detect pancreatic cancer earlier. This would allow for earlier treatment and better patient results. It also encourages more research to refine and confirm these methods for use in real-world medical settings.[Far+24] Farhangnia et al. (2024) present a comprehensive overview of existing and prospective immunotherapy strategies in pancreatic cancer, focusing especially on pancreatic ductal adenocarcinoma (PDAC), recognized as one of the deadliest cancers globally. Even with improvements in surgery, chemotherapy, and radiotherapy, overall survival rates are still low, and conventional immunotherapies that have transformed treatment for other cancers have shown only restricted effectiveness in PDAC. This is mainly due to the distinctive and strongly immunosuppressive tumor microenvironment (TME), noted for its dense desmoplastic stroma, limited vascularization, hypoxia, and a high presence of immunosuppressive cells like tumor-associated macrophages (TAMs), myeloid-derived suppressor cells (MDSCs), and regulatory T cells. These characteristics serve as a physical barrier to drug and immune cell penetration while also actively inhibiting effector T-cell function, resulting in an environment that resists immune eradication. Additionally, the generally low mutational burden in PDAC leads to a reduced number of neoantigens, restricting the adaptive immune system's ability to recognize tumor cells. The writers classify immunotherapy approaches into various supportive methods. Immune checkpoint inhibitors (ICIs), including PD-1/PD-L1 and CTLA-4 inhibitors, have demonstrated restricted activity as standalone treatments but are progressively undergoing testing alongside chemotherapy, radiotherapy, or stromal-targeting agents to improve their effectiveness. Developing engineered cell-based therapies, especially CAR-T and CAR-NK cells targeting PDAC-related antigens, faces challenges due to the adverse TME. A different encouraging approach includes therapeutic cancer vaccines, varying from peptide-based to neoantigen-targeted types, designed to enhance long-lasting antigen-specific responses. Oncolytic viruses serve as an extra approach, not just by directly destroying tumor cells but also by liberating tumor-associated antigens and boosting local immune response. Concurrent initiatives aim at reprogramming the immunosuppressive myeloid and stromal elements of PDAC, through either the depletion of suppressive cell populations or the transition towards pro-inflammatory phenotypes. These methods are further reinforced by cutting-edge drug delivery techniques that can improve therapeutic penetration through the thick stroma, like nanoparticle carriers. Crucially, there is little chance that PDAC patients will have significant clinical gain from monotherapy using any of these strategies. Rather, the most promising approach seems to be combinatorial regimens that combine immunotherapy with other immune-based approaches or with traditional treatments. Furthermore, as treatment outcomes are greatly impacted by heterogeneity in tumor biology and immune responses, the discovery of predictive biomarkers for patient stratification is also essential. Since existing preclinical models frequently fall short of capturing the intricacy of stromal-immune interactions seen in patients, the authors therefore stress the need for new representative models that faithfully replicate the actual PDAC milieu. Farhangnia et al. (2024) provide an extensive review of current and future immunotherapeutic approaches in pancreatic cancer, with particular emphasis on pancreatic ductal adenocarcinoma (PDAC),which remains one of the most lethal malignancies worldwide. Despite advances in surgery,chemotherapy, and radiotherapy, overallsurvival rates remain poor, and traditional immunotherapies that have revolutionized treatment in other cancers have demonstrated only limited success in PDAC.This is primarily attributed to the unique and highly immunosuppressive tumor microenvironment (TME),which is characterized by a dense desmoplastic stroma, restricted vascularization, hypoxia, and an abundance of immunosuppressive cells such as tumor-associated macrophages (TAMs), myeloid-derived suppressor cells (MDSCs), and regulatory T cells. These features not only act as a physical barrier to drug and immune cell infiltration but also actively suppress effector T-cell activity, thereby creating an environment resistant to immune clearance. Moreover, the typically low mutational burden of PDAC results in fewer neoantigens, which limits the recognition of tumor cells by the adaptive immune system.The authors categorize immunotherapeutic interventions into multiple complementary strategies.Immune checkpoint inhibitors (ICIs), such as PD-1/PD-L1 and CTLA-4 inhibitors, have shown limited single-agent activity but are increasingly being tested in combination with chemotherapy, radiotherapy, or stromal-targeting agents to enhance their efficacy. Engineered cell-based therapies, particularly CAR-T and CAR-NK cells directed against PDAC-associated antigens, are also being developed, though their success is still constrained by the hostile TME. Another promising avenue involves therapeutic cancer vaccines, ranging from peptide-based to neoantigen-targeted vaccines, which aim to boost durable antigen-specific responses. Oncolytic viruses represent an additional strategy, not only by directly lysing tumor cells but also by releasing tumor-associated antigens and enhancing local immune activation. Parallel efforts focus on reprogramming the immunosuppressive myeloid and stromal components of PDAC, either by depleting suppressive cell populations or by shifting them toward pro-inflammatory phenotypes. These approaches are further supported by novel drug delivery systems,such as nanoparticle carriers, which can enhance therapeutic penetration through the dense stroma.Importantly, the review highlights that monotherapy with any of these approaches is unlikely to achieve meaningful clinical benefit in PDAC. Instead, combinatorial regimens that integrate immunotherapy with conventional treatments or with other immune-based interventions appear to be the most promising strategy. In addition, the identification of predictive biomarkers for patient stratification remains crucial, as heterogeneity in tumor biology and immune responses significantly affects treatment outcomes. The authors also emphasize the need for more representative preclinical models that accurately mimic the human PDAC microenvironment, since current models often fail to capture the complexity of stromal-immune interactions observed in patients.Overall, Farhangnia et al. (2024) argue that while immunotherapy alone has thus far failed to achieve breakdown in pancreatic cancer, the integration of multi-pronged approaches—targeting both tumor cells and the surrounding immunosuppressive microenvironment—offers a rational path forward. Their work underlines the importance of combinatorial strategies, stromal and myeloid targeting, and personalized immunotherapy guided by biomarkers, positioning these as essential directions for the next generation of pancreatic cancer research and treatment development.

# Chapter 3

# Data Exploration

## 3.1 Data Collection

This chapter serves as a comprehensive account of the methodology employed to investigate pancreatic cancer detection using Computed Tomography (CT) scan images, with a distinct and rigorous emphasis on data visualization techniques. The structured approach is designed for maximum clarity and scientific reproducibility, beginning with a detailed description of the source dataset and systematically progressing through the techniques applied for preprocessing, visualization, and subsequent analysis. The specific methods chosen—from image manipulation to final visual representation—are critically designed not merely to display data, but to effectively highlight the morphological differences between normal pancreatic structures and tumorous regions, ensuring the derived visual insights contribute meaningfully to a deeper, more intuitive understanding of the disease's characteristics.The foundation of this research rests upon a comprehensive, open-source dataset of anonymized CT images, which provides a robust and ethical source of material for study, totaling 1,411 samples. This collection has been systematically and intentionally divided to create a controlled environment for testing: a large training set consisting of 999 images (421 normal and 578 tumour) is used for model development, while an isolated testing set comprising 412 images (225 normal and 187 tumour) is strictly reserved for unbiased evaluation.Seen in figure 3.1. This clear and predetermined partitioning of the data provides a robust basis not only for direct visual comparison between normal and diseased cases but also for subsequent algorithmic analysis. Ultimately, this foundational structure ensures that the techniques employed for preprocessing, visualization, and analysis yield transparent and reproducible visual insights that directly contribute to a better understanding of the morphological hallmarks that are critical for the clinical diagnosis and effective treatment planning of pancreatic cancer.

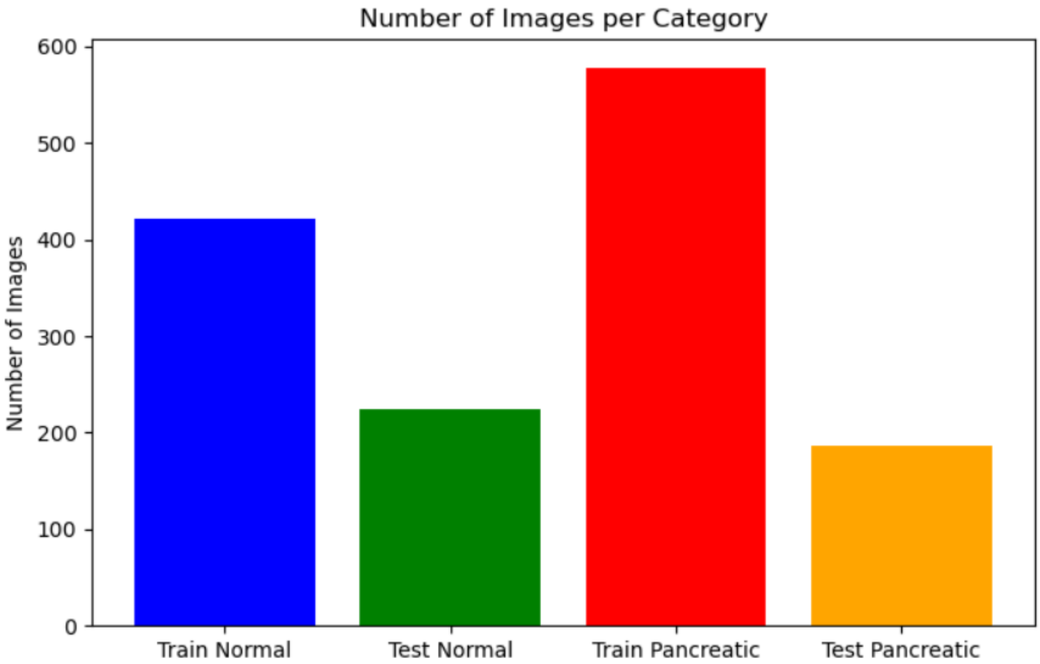



Figure 3.1: Distribution of Pancreatic Cancer images by Computed Tomography

## 3.2 Pre-processing

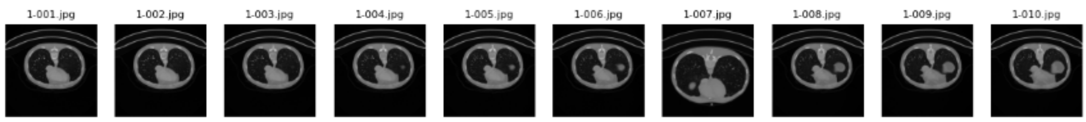
The dataset utilized in this study was sourced from Kaggle by [Pon23], a widely used platform for publicly available datasets, and consists of images of both normal pancreatic tissue and pancreatic tumors. The dataset was pre-divided into training and testing sets to ensure proper evaluation of the models, allowing the training phase to occur on one set while assessing performance on a completely separate testing set. This division helps prevent overfitting and provides a more accurate measure of how well the models generalize to unseen data.

Originally, the images were captured at a resolution of 512×512 pixels, but for visualization, processing efficiency, and consistency, all images were resized to 224×224 pixels as seen in figure 3.2. This resizing ensures uniformity across all images, which is particularly important when feeding the images into classification models or performing any comparative analysis. To focus on intensity-based patterns relevant for distinguishing normal from tumor tissue, all images were converted to grayscale. Converting to a single channel reduces computational complexity without losing critical information that may contribute to accurate classification. The images are stored in the widely used JPG format, which balances storage efficiency with image quality. Preliminary examination of the dataset revealed some minor variations in brightness, contrast, and texture across the images, likely due to differences in image acquisition methods or inherent tissue properties. These variations are important to note as they can influence the performance of image processing and machine learning models, and appropriate preprocessing steps will address them. In terms of dataset composition, the number of images in each category is relatively balanced, although

```python
import os
import pandas as pd
from PIL import Image
import matplotlib.pyplot as plt

image_folder = r"C:\data\DATASET"

files = []
for root, dirs, filenames in os.walk(image_folder):
    for f in filenames:
        if f.lower().endswith((".jpg", ".jpeg")):
            files.append(os.path.join(root, f))

print(f"Found {len(files)} images.")

 #Create DataFrame
df = pd.DataFrame({"filepath": files})
df["filename"] = df["filepath"].apply(lambda x: os.path.basename(x))
df["width"] = None
df["height"] = None
df["corrupted"] = False
df['class'] = df['filepath'].apply(lambda x: os.path.basename(os.path.dirname(x)))



#  Check for corrupted images and get dimensions
for idx, path in enumerate(df["filepath"]):
    try:
        img = Image.open(path)
        img.verify()  # Check if image is corrupted
        img = Image.open(path)  # Reopen to get size
        df.at[idx, "width"], df.at[idx, "height"] = img.size
    except:
        df.at[idx, "corrupted"] = True

# Remove corrupted images if any
df = df[df["corrupted"] == False].reset_index(drop=True)
print(f"Usable images: {len(df)}")

#  Display first 10 images inline
fig, axes = plt.subplots(1, 10, figsize=(15,5))
for i, ax in enumerate(axes):
    img = Image.open(df.loc[i, "filepath"]).convert("L")
    ax.imshow(img,cmap="gray")
    ax.set_title(df.loc[i, "filename"], fontsize=8)
    ax.axis("off")
plt.show()


df.head()
```

Figure 3.2: Visualization of Original Pancreatic and Normal Images from the Dataset

```
Found 1411 images.
Usable images: 1411
```




| | filepath | filename | width | height | corrupted | class |
|---|---|---|---|---|---|---|
| 0 | C:\data\DATASET\test\test\normal\1-001.jpg | 1-001.jpg | 512 | 512 | False | normal |
| 1 | C:\data\DATASET\test\test\normal\1-002.jpg | 1-002.jpg | 512 | 512 | False | normal |
| 2 | C:\data\DATASET\test\test\normal\1-003.jpg | 1-003.jpg | 512 | 512 | False | normal |
| 3 | C:\data\DATASET\test\test\normal\1-004.jpg | 1-004.jpg | 512 | 512 | False | normal |
| 4 | C:\data\DATASET\test\test\normal\1-005.jpg | 1-005.jpg | 512 | 512 | False | normal |

Figure 3.3: General overview of the first 10 images in their original size

slight differences exist between the normal and tumor classes. The training set contains the majority of images for both classes, while the testing set provides a smaller but representative sample to evaluate model performance. This structured distribution ensures that models are trained on sufficient data while maintaining a reliable assessment of predictive accuracy. Overall, this standardized and carefully organized dataset forms the foundation for all subsequent steps in the study, including preprocessing, feature extraction, model training, and evaluation. Understanding these characteristics is essential for ensuring that the applied methods are appropriate and that the results obtained are meaningful and reliable. Before training, each image underwent several pre-processing steps to ensure consistency and to transform the visual information into a numerical format suitable for computational models.

1. Image resizing : All images were resized from their original resolution (512 × 512 pixels) to 224 × 224 pixels. This resizing reduces computational cost while maintaining sufficient structural information for classification tasks.Resizing is a crucial, early-stage data preparation step where all 512×512 images are physically reduced to 224×224 pixels, a task often accomplished using tools like Pillow and orchestrated by file locators like Glob. This downsampling serves to reduce computational cost by significantly decreasing the data volume and ensures the images possess sufficient structural information by complying with the standard input size for efficient transfer learning with established CNN architectures. Figure 3.4

$$I' = \mathrm{Resize}(I,\ 224 \times 224)$$

2. Flattening into Numerical Vectors : Each 2D scale grayscale image size 224 X 224 was flattened into a 1D feature vector of length 224*224 = 50,176. . This transformation enables direct representation of images in a numerical matrix, where each row corresponds to an image and each column corresponds to a pixel intensity.Resizing is performed directly on the raw image files, whereas flattening denotes a conceptual transformation that occurs later in the processing workflow, generally immediately prior to inputting the data into a conventional neural network layer or employing it within feature-based classification techniques.

$$x = \mathrm{Flatten}(I) \in \mathbb{R}^{50176}$$

3. Data structure : Each category was represented as a shape matrix (N, 50176), where N is the number of images. The 50,176 columns represent the features extracted from each image. Since the images were pre-processed to be 224×224 grayscale and then flattened, 224×224=50,176 pixels, meaning each column holds the intensity value of a specific pixel location across all images in that category.

```
import glob
from PIL import Image
import numpy as np
import matplotlib.pyplot as plt

train_normal_path = r"C:\data\DATASET\train\train\normal\*"
train_pancreatic_path = r"C:\data\DATASET\train\train\pancreatic_tumor\*"
test_normal_path = r"C:\data\DATASET\test\test\normal\*"
test_pancreatic_path = r"C:\data\DATASET\test\test\pancreatic_tumor\*"


def load_images(path, size=(224, 224), grayscale=True):
    images = []
    for file in glob.glob(path):
        if grayscale:
            img = Image.open(file).convert('L')  # grayscale
        else:
            img = Image.open(file).convert('RGB')  # color
        img = img.resize(size)
        img_array = np.array(img)
        if grayscale:
            img_array = img_array.flatten()      # flatten for grayscale
        images.append(img_array)
    return np.array(images)

train_normal = load_images(train_normal_path)
train_pancreatic = load_images(train_pancreatic_path)
test_normal = load_images(test_normal_path)
test_pancreatic = load_images(test_pancreatic_path)
print("Train Normal:", train_normal.shape)
print("Train Pancreatic:", train_pancreatic.shape)
print("Test Normal:", test_normal.shape)
print("Test Pancreatic:", test_pancreatic.shape)

categories = ['Train Normal', 'Test Normal', 'Train Pancreatic', 'Test Pancreatic']
counts = [len(train_normal), len(test_normal), len(train_pancreatic), len(test_pancreatic)]

plt.figure(figsize=(8,5))
plt.bar(categories, counts, color=['blue', 'green', 'red', 'orange'])
plt.ylabel('Number of Images')
plt.title('Number of Images per Category')
plt.show()
```

Figure 3.4: Resizing and Flattening of images

```
Train Normal: (421, 50176)
Train Pancreatic: (578, 50176)
Test Normal: (225, 50176)
Test Pancreatic: (187, 50176)
```

Figure 3.5: Data Structure

```
import numpy as np
normal_test_images = [img.flatten()for img in normal_test_images]
pancreatic_tumor_test_images = [ img.flatten()for img in pancreatic_tumor_test_images]
all_test_images = normal_test_images + pancreatic_tumor_test_images
data_array = np.array(all_test_images)
print(f"final array has a shape{data_array.shape}.")
```

```
final array has a shape(412, 50176).
```

Figure 3.6: Pancreatic test set and Normal test set

In this work, I have deliberately chosen to focus on the test sets of the already-partitioned dataset for both algorithm evaluation and visualization purposes. This strategic decision stems from the central principle in machine learning that reliable model performance must be assessed on data not seen during training. The test set represents such unseen data and, as a result, provides the most faithful and unbiased estimate of a model's ability to generalize to real-world scenarios. By contrast, the training set, while indispensable for parameter optimization, is highly vulnerable to overfitting: the model may achieve artificially high accuracy on training examples simply because it has memorized their features, yet this performance often deteriorates when exposed to entirely new instances. Consequently, evaluation on the training set risks producing overly optimistic and misleading conclusions regarding model effectiveness.By isolating the test set data structure—the matrix of shape (N,50176) that represents the flattened 224×224 images—I ensure that my model's performance metrics and all subsequent visualizations truly reflect its reliable, real-world effectiveness.

# Chapter 4

# Classification Algorithm

This chapter provides a systematic and comprehensive account of the classification and analytical techniques applied to the CT scan dataset for pancreatic cancer detection. Following the preprocessing and initial visualization described in the previous chapter, this section delves deeper into both statistical and algorithmic analyses, with a focus on uncovering meaningful patterns and ensuring accurate classification of normal and tumorous pancreatic structures. The methodology adopted in this chapter is designed to progress logically from exploratory data characterization to advanced machine learning-based classification, thereby ensuring clarity, reproducibility, and clinical relevance. The analytical process begins with matrix correlation, which examines the relationships between flattened image matrices in both the training and test sets. This step provides insight into the underlying structure of the dataset and helps identify patterns that may distinguish normal from tumorous images. Next, pixel intensity graphs are plotted for both training and test sets, offering a visual understanding of the distribution of grayscale values across different classes. Complementing this, the distribution of average brightness is analyzed specifically for the test set, providing a quantitative measure of overall luminance differences between healthy and pathological tissues. To further assess image similarity, the Structural Similarity Index (SSIM) is employed, allowing for an objective evaluation of perceptual differences and highlighting subtle morphological changes relevant to tumor detection. Building upon these visualization and feature-analysis steps, the chapter then introduces classification methodologies aimed at translating these insights into predictive modeling. A Support Vector Machine (SVM) is first applied to extract class boundaries based on the features derived from the images. Subsequently, XGBoost is utilized in two contexts: first, applied on SVM-extracted features to enhance classification performance through ensemble learning, and second, applied to features derived from a pretrained AlexNet model, leveraging deep learning representations for improved detection accuracy. By integrating both classical machine learning and deep learning approaches, this chapter ensures a robust framework for assessing model performance and highlights the complementary strengths of these methods. Collectively, these seven steps—from correlation matrices and pixel-level analysis to advanced classification—form a cohesive methodology that combines visualization, quantitative assessment, and predictive modeling. This structured approach not only facilitates a clear understanding of the morphological differences between normal and tumorous pancreatic tissues but also ensures that the results are reliable, reproducible, and applicable for practical clinical decision-making.

## 4.1 Structural Similarity Index Measure

The Structural Similarity Index (SSIM) has become a favored metric in picture quality assessment and comparison analysis. By simulating human visual experience, SSIM quantifies image degradation in contrast to conventional fidelity measurements like Mean Squared Error (MSE) or Peak Signal-to-Noise Ratio (PSNR), which calculate point-by-point pixel differences. This is accomplished by comparing local patterns of pixel intensities between a reference and a deformed image, taking into account evaluations of brightness, contrast, and structure. Because it is observably more consistent with subjective human assessments of image fidelity, this method offers a more semantically meaningful measure of visual similarity and is especially useful for assessing how well image processing algorithms such as compression, restoration, and reconstruction perform. The SSIM score was computed using the skimage.metrics module in Python. The implementation involved loading both the normal and tumour images in grayscale format. The data.range was explicitly set to 255 to ensure accurate normalization across the 8-bit pixel intensity spectrum. Crucially, the function was executed to return the SSIM Difference Map, which provides a spatial visualization of the structural dissimilarities between the two images. For clearer analysis, a red bounding box was programmatically added to the tumour image and the difference map to visually highlight and localize the direction of the tumour or the anomaly. This map was then meticulously analyzed to visually confirm whether the areas of lowest structural similarity (highest difference) correlate directly with the radiologically identified tumour regions, thus providing a crucial initial visual and quantitative confirmation of the structural challenge faced by the subsequent classification models.

```python
NORMAL_FILE    = r"C:\data\DATASET\test\test\normal\1-054.jpg"
TUMOR_FILE     = r"C:\data\DATASET\test\test\pancreatic_tumor\1-043.jpg"

DATA_RANGE = 255
BOX_X = 150
BOX_Y = 150
BOX_WIDTH = 220
BOX_HEIGHT = 200

# --- 2. Load Images ---
normal = cv2.imread(NORMAL_FILE, cv2.IMREAD_GRAYSCALE)
tumor  = cv2.imread(TUMOR_FILE, cv2.IMREAD_GRAYSCALE)

print("\n--- Running Targeted SSIM Comparison ---")
print("Normal Image:", NORMAL_FILE)
print("Tumor Image:", TUMOR_FILE)


score, diff = ssim(normal, tumor, full=True, data_range=DATA_RANGE)
print(f"SSIM Score (Normal vs. Tumor): {score:.4f}")


fig, axes = plt.subplots(1, 3, figsize=(15, 5))
plt.rcParams['font.size'] = 10

axes[0].imshow(normal, cmap='gray')
axes[0].set_title(f"1. Normal Slice\n(File: 1-054.jpg)")
axes[0].axis('off')


axes[1].imshow(tumor, cmap='gray')
axes[1].set_title(f"2. Tumor Slice\n(File: 1-043.jpg) - Boxed Anomaly")
axes[1].axis('off')

rect_tumor = patches.Rectangle((BOX_X, BOX_Y), BOX_WIDTH, BOX_HEIGHT,
                               linewidth=2, edgecolor='red', facecolor='none')
axes[1].add_patch(rect_tumor)


im = axes[2].imshow(diff, cmap='viridis', vmin=0, vmax=1)
axes[2].set_title(f"3. SSIM Difference Map\nScore: {score:.4f} - Boxed Difference")
axes[2].axis('off')


rect_diff = patches.Rectangle((BOX_X, BOX_Y), BOX_WIDTH, BOX_HEIGHT,
                              linewidth=2, edgecolor='red', facecolor='none')
axes[2].add_patch(rect_diff)


plt.suptitle("Targeted SSIM Analysis: Structural Difference Bounding Box", y=1.02, fontsize=14)
plt.tight_layout()
plt.show()
```

Figure 4.1: SSIM performed on Pancreatic test set and Normal test set

## 4.2 Feature Space Analysis

In image classification, analyzing the feature space gives a mathematical way to show images. Each pixel or feature in the image becomes a part of a space with many dimensions. An image is made up of numbers that show brightness or color, and the feature space is created by turning these numbers into vectors.

$$\mathbf{f} = [f_1, f_2, \ldots, f_n]$$

**f** represents a statistical or structural property of the image such as pixel intensity, brightness, or texture.By showing images in this feature space, we can study patterns, how well different groups separate, and how they spread out before using classification methods. For example, pixel intensity shows how light and dark areas are spread in the image, and brightness shows the average light levels. Both are simple ways to describe the visual details of medical images and help spot small differences between tissues. So, looking at this feature space is the first step in understanding these images and linking them to more advanced ways of sorting them into categories.

### 4.2.1 Pixel Intensity Distribution

To better understand how different classes can be separated in the feature space, I looked at the pixel brightness levels in images of pancreatic tissue compared to normal body tissue. Because the dataset is big, we picked 20 random test images from each group to show the differences clearly. An image I(x,y) in grayscale can be modeled as a matrix of intensity values where x and y denote the pixel coordinates and the intensity ranges from 0(black) to 256(white).

$$H(k) = \sum_{x=1}^{M} \sum_{y=1}^{N} \delta(I(x, y) - k)$$

This distribution shows how pixel values are spread out across different shades of gray, which helps compare the texture and contrast between pancreatic and non-pancreatic tissues. Differences in these distributions can show small structural details, making intensity histograms a helpful tool for classification at a basic level. Here normal test images and pancreatic tumor test images create two lists, each containing 50 arrays. Each array represents a simulated 224×224 grayscale image, with pixel values randomly chosen between 0 (black) and 256 (white). Due to the large size of the actual image dataset (where 224×224 pixels per image would quickly lead to unmanageable memory usage during preliminary analysis), the code intentionally uses randomly generated data for demonstration.

```python
import matplotlib.pyplot as plt
import numpy as np

normal_test_images = [np.random.randint(0,256, size=(224,224)) for _ in range(50)]
pancreatic_tumor_test_images = [np.random.randint(0, 256, size=(224,224)) for _ in range(50)]

all_normal_pixels = np.concatenate(normal_test_images, axis=None)
all_tumor_pixels = np.concatenate(pancreatic_tumor_test_images, axis=None)

data = [all_normal_pixels, all_tumor_pixels]
labels = ['Normal Body Images', 'Pancreatic Tumor Images']

plt.figure(figsize=(10, 7))
bp = plt.boxplot(data, vert=True,labels=labels, patch_artist = True)
colors = ['green','blue']
for patch,color in zip(bp['boxes'],colors):
    patch.set_facecolor(color)

#title and labels
plt.title('Pixel Intensity Distribution: Normal vs. Tumor Images', fontsize=10)
plt.ylabel('Pixel Intensity', fontsize= 7)
plt.xlabel('Image Type', fontsize= 7)
plt.grid(True, alpha=0.7)
plt.show()
```

Figure 4.2: Pixel Intensity between Pancreatic test set and Normal test set

```
import matplotlib.pyplot as plt
import numpy as np
from glob import glob
from PIL import Image

normal_test_files = glob(r"C:\data\DATASET\test\test\normal\*")
pancreatic_tumor_test_files = glob(r"C:\data\DATASET\test\test\pancreatic_tumor\*")
normal_averages = [np.array(Image.open(file).convert('L')).mean() for file in normal_test_files]
pancreatic_tumor_averages = [np.array(Image.open(file).convert('L')).mean() for file in pancreatic_tumor_test_files]
plt.figure(figsize=(10, 6))
plt.hist(normal_averages, bins=20, color='blue', alpha=0.5, label='Normal Images')
plt.hist(pancreatic_tumor_averages, bins=20, color='red', alpha=0.5, label='Pancreatic Tumor Images')
plt.title('Distribution of Average Image Brightness')
plt.xlabel('Average Pixel Intensity per Image')
plt.ylabel('Number of Images')
plt.legend()
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```

Figure 4.3: Image brightness

### 4.2.2 Image Brightness Distribution

In addition to analyzing intensity, I also looked at how the average brightness of images was spread out across the test set for both categories. Brightness gives a general idea of how light or dark an image is, and it can be calculated by finding the average intensity.

$$B = \frac{1}{M \times N} \sum_{x=1}^{M} \sum_{y=1}^{N} I(x, y)$$

The image brightness distribution shows how the average brightness changes across all the images in the dataset. It reflects the overall light levels in the images rather than the brightness of individual pixels. When we compare the brightness patterns of images that show the pancreas with those that don't, we can see if there are clear differences in the overall light levels between the two groups. This helps in distinguishing between the two types early on and can also be used as an extra feature to help classify images.

```python
import pandas as pd
import numpy as np
data_array = np.random.rand(20, 20)
df = pd.DataFrame(data_array)
correlation_matrix = df.corr()
correlation_df = pd.DataFrame(correlation_matrix)

# This line ensures the full matrix is printed.
np.set_printoptions(threshold=np.inf)

print("Correlation Matrix Shape (Rows, Columns):")
print(correlation_df.shape)
print("\nFull Correlation Matrix:")
print(correlation_df)
```

Figure 4.4: Numerical Brightness

## 4.3 Correlation Analysis of Transformed Image Data

The basis of this analysis is turning raw image data into a form that computers can easily work with. Since our dataset includes a variety of images, the first important step was converting each image into a numerical matrix. This process makes it easier to apply algorithms and also helps in creating a clear, visual way to understand the structure of features within the images. By turning images into matrices, we can better explore how different features relate to each other, which leads to the next step of analyzing correlations. While the human eye can perceive patterns and anomalies in these images, computational analysis requires a structured numerical representation that algorithms can process mathematically. This numerical representation unlocks powerful analytical capabilities that would be impossible with raw image files. Matrix operations, statistical analyses, machine learning algorithms, and visualization techniques all operate on numerical data, making the matrix format essential for any computational approach to medical image analysis. In the specific context of pancreatic cancer detection, where early diagnosis is critically important but extremely challenging, the ability to quantitatively analyze subtle imaging features can make the difference between timely intervention and late-stage diagnosis. The correlation analysis that follows the image-to-matrix conversion serves as both a data exploration tool and a quality assurance mechanism. By examining how different regions or features within the images relate to one another statistically, we gain insights into the underlying structure of the data, identify potential redundancies or dependencies, and prepare for more sophisticated modeling approaches. In medical imaging, understanding these correlations can reveal anatomical consistency, technical artifacts, or pathological patterns that inform subsequent analysis steps.

### 4.3.1 Technical Implementation

The code implementation utilizes Python's powerful data science libraries—pandas and numpy—to handle the numerical transformations and statistical computations. The process begins with the generation of a random numerical array using np.random.rand(20,20), which creates a 20×20 matrix of random floating-point values between 0 and 1. In the actual implementation with image data, this array would represent the pixel intensity values extracted from the images, where each cell in the matrix corresponds to a specific pixel location, and the numerical value represents the grayscale intensity or a specific color channel value. The conversion of this numerical array into a pandas DataFrame `df = pd.DataFrame(data_array)` is a crucial step that provides several advantages. DataFrames offer a structured, tabular representation of the data with labeled rows and columns, making it significantly easier to manipulate, analyze, and visualize the information. This structure is particularly beneficial when dealing with image data converted to numerical form, as it allows for intuitive indexing, slicing, and statistical operations that would be more cumbersome with raw numpy arrays. The next step involves a correlation matrix which is a square matrix that displays the correlation coefficients between all pairs of features (columns) in the dataset. Each element in this matrix represents the Pearson correlation coefficient, which measures the linear relationship between two variables, ranging from -1 to +1. A correlation coefficient of +1 indicates a perfect positive linear relationship, -1 indicates a perfect negative linear relationship, and 0 suggests no linear relationship. In the context of image analysis, the correlation matrix reveals how different pixel positions or image features relate to one another. For instance, if we're analyzing medical images such as CT scans of the pancreas, adjacent pixels often show high correlation because nearby regions typically share similar intensity values.

## 4.4 Support Vector Machine

Support Vector Machine (SVM) is a powerful supervised machine learning algorithm that has become one of the most widely used techniques in medical image analysis and disease classification. At its core, the Support Vector Machine (SVM) finds the best possible decision boundary (hyperplane) to separate data by maximizing the margin between the different classes. The algorithm's effectiveness and name come from the support vectors: the few crucial data points closest to the boundary that completely determine its optimal position. This selective approach, ignoring all but the most critical boundary points, is key to SVM's high efficiency and generalization power. The Support Vector Machine (SVM) stands apart from many machine learning models because its design is rooted in statistical learning theory, specifically the concept of Structural Risk Minimization (SRM). While most algorithms simply strive to minimize errors on the training data, SVM pursues a critical balance: it seeks a decision boundary that not only correctly classifies the examples it was trained on but also generalizes well to any new data it sees. This careful, built-in mechanism against complexity provides regularization, making the SVM highly resistant to overfitting. This makes it an especially valuable tool in fields like medical imaging, where obtaining large, labeled datasets is often expensive and challenging.

### 4.4.1 Technical Implementation

The implementation begins with a systematic approach to loading and preparing image data for machine learning analysis. The load_images() function serves as the cornerstone of the data preparation pipeline, accepting three critical parameters: the directory path containing the images, a label identifier to associate with those images, and a standardized size tuple (224, 224) that defines the uniform dimensions to which all images will be resized. The use of *glob* function provides a powerful mechanism for dynamically discovering all image files within the specified directory, eliminating the need for manual file enumeration and making the code adaptable to datasets of varying sizes. This approach ensures scalability and flexibility, as the function can process any number of images without modification. This grayscale conversion is a critical preprocessing step that reduces the three-channel RGB color representation to a single-channel intensity representation. While this may seem like a loss of information, grayscale conversion offers several important advantages in medical image analysis, it reduces computational complexity by eliminating color channel dimensions, standardizes the representation across images that may have been acquired with varying color profiles or display settings, and focuses the analysis on intensity patterns and textures rather than color information—which is often more relevant for diagnostic features in CT scans and other medical imaging modalities. While flattening destroys the explicit 2D spatial structure of the image, the consistent mapping from spatial coordinates to feature indices is preserved across all images through their ordering in the flattened array, allowing the SVM to still capture spatial patterns by learning relationships between feature dimensions that correspond to spatially related pixels.The resize() operation transforms all images to exactly 224×224 pixels, ensuring dimensional consistency required for machine learning algorithms. he implementation constructs training and testing datasets through multiple load_images() calls, processing normal and tumor images separately. Binary labels are assigned: 0 for normal tissue and 1 for tumor tissue, following standard medical classification conventions where 1 indicates disease presence. Concatenation operations combine these collections using X_train_normal + X_train_pancreatic to create unified feature matrices where each row represents one flattened image (50,176 pixel values) and each column represents a specific pixel position. The test_size=0.3 parameter allocates 30% of data for testing and 70% for training for accurate evaluation. Before SVM training, StandardScaler() applies feature standardization, a mathematically critical preprocessing step. The scaler computes the mean and standard deviation of each feature across all training images, then transforms features to have zero mean and unit variance using the formula:

$$z = \frac{x - \mu}{\sigma}$$

where x is the original pixel value. Standardization is crucial for SVMs because the algorithm's decisions are based on distance. Without standardization, pixels with naturally higher variance would dominate distance calculations and receive disproportionate importance regardless of their actual diagnostic value.

- **F1 Score**
  The F1 score provides a balanced evaluation metric particularly appropriate for medical classification tasks, especially when dealing with potentially imbalanced datasets or when both types of classification errors carry significant clinical consequences. The F1 score is calculated as the harmonic mean of precision and recall, two fundamental performance metrics that capture different aspects of classification quality.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

$$\text{Recall (Sensitivity)} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

  In pancreatic cancer detection, both missing tumors and false alarms have serious consequences for patients, so the F1 score provides a balanced measure that reflects real clinical needs and helps determine if the model trained on this dataset is suitable for practical use.

```
import glob
from PIL import Image
import numpy as np
from sklearn.svm import SVC
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import f1_score, classification_report
from sklearn.model_selection import train_test_split

train_normal_path = r"C:\data\DATASET\train\train\normal\*"
train_pancreatic_path = r"C:\data\DATASET\train\train\pancreatic_tumor\*"
test_normal_path = r"C:\data\DATASET\test\test\normal\*"
test_pancreatic_path = r"C:\data\DATASET\test\test\pancreatic_tumor\*"

def load_images(path, label, size=(224,224)):
    data, labels = [], []
    for file in glob.glob(path):
        img = Image.open(file).convert("L")
        img = img.resize(size)
        data.append(np.array(img).flatten())
        labels.append(label)
    return data, labels

X_train_normal, y_train_normal = load_images(train_normal_path, 0)
X_train_pancreatic, y_train_pancreatic = load_images(train_pancreatic_path, 1)
X_test_normal, y_test_normal = load_images(test_normal_path, 0)
X_test_pancreatic, y_test_pancreatic = load_images(test_pancreatic_path, 1)

X = np.array(X_train_normal + X_train_pancreatic + X_test_normal + X_test_pancreatic)
y = np.array(y_train_normal + y_train_pancreatic + y_test_normal + y_test_pancreatic)


X_train, X_eval, y_train, y_eval = train_test_split(X, y, test_size=0.3, random_state=42, stratify=y)


scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_eval = scaler.transform(X_eval)

clf = SVC(kernel='linear',probability = True , random_state=42)
clf.fit(X_train, y_train)

y_pred = clf.predict(X_eval)

f1 = f1_score(y_eval, y_pred, average='binary')
print("F1 Score:", f1)

print("\nClassification Report:\n", classification_report(y_eval, y_pred))
```

Figure 4.5: Support Vector Machine

## 4.5 Ensemble Learning with XGBoost for Image Classification

XGBoost (eXtreme Gradient Boosting) is a popular and powerful machine learning algorithm that has been widely adopted for classification tasks, including image classification. It belongs to the ensemble learning family, which means it combines multiple weak learners (typically decision trees) to create a stronger, more accurate model. The key principle behind XGBoost is sequential learning—each new tree in the ensemble focuses on correcting the mistakes made by the previous trees, gradually improving the overall prediction accuracy. In image classification tasks, XGBoost is particularly useful when working with extracted features rather than raw pixel values. While deep learning models like convolutional neural networks can directly process images, XGBoost excels at learning patterns from pre-extracted feature representations. This makes it an ideal choice for building on top of feature extraction methods, whether those features come from traditional machine learning classifiers or deep learning architectures. There are several reasons why XGBoost is effective for image classification. First, it can handle high-dimensional feature spaces efficiently, which is important when dealing with the numerous features extracted from images. Second, it includes regularization techniques that help prevent overfitting, ensuring the model generalizes well to unseen data. Third, XGBoost provides insights into feature importance, allowing us to understand which features contribute most to the classification decisions. In this study, XGBoost was applied in two configurations: first, as a classifier on top of Support Vector Machine (SVM) features, and second, using features extracted from the AlexNet deep learning architecture. This approach combines the strengths of different methods—using established feature extraction techniques and then applying XGBoost's powerful ensemble learning to refine the classification results and potentially achieve better performance than the base classifiers alone.

### 4.5.1 Extreme Gradient Boosting performed on Support Vector Machine

The combination of XGBoost with Support Vector Machines creates a two-step classification method that leverages the strengths of both traditional machine learning and ensemble techniques. In this approach, SVM acts as the first classifier, and then XGBoost is used as a second-level classifier to improve the final predictions. The process works in two main stages. First, an SVM classifier with a linear kernel is trained on the image features extracted from the dataset. Rather than just using the SVM's final predictions (whether an image is normal or has a pancreatic tumor), we also extract the probability scores that the SVM generates. These probability scores tell us how confident the SVM is about its prediction for each image. For example, a probability of 0.9 for the tumor class means the SVM is very confident that the image shows a tumor, while a probability of 0.55 suggests the SVM is less certain. In the second stage, these probability scores become the input features for XGBoost. Instead of looking at the original image features, XGBoost learns to make decisions based on the SVM's confidence levels. Essentially, XGBoost is learning to interpret and refine what the SVM has already learned. This creates a form of meta-learning, where one model learns from another model's understanding of the data. By using XGBoost on top of SVM, we give the model a chance to learn from patterns in the SVM's confidence scores and make better decisions, especially for those uncertain cases. For instance, if the SVM consistently gives probability scores around 0.6 for certain types of images that are actually tumors, XGBoost can learn this pattern and adjust its decision-making accordingly. This adaptive learning is something that a single SVM classifier cannot easily achieve on its own. The main advantage of this hybrid method is that it combines SVM's strength in creating well-defined classification boundaries with XGBoost's ability to handle complex patterns through ensemble learning. SVM provides a solid initial classification based on mathematical optimization principles—it finds the hyperplane that best separates the two classes with maximum margin. Then, XGBoost acts as a "refiner" that can correct mistakes or improve uncertain predictions by building multiple decision trees that focus on the errors made in previous iterations. Another important benefit is feature space transformation. The SVM probability outputs create a simplified, two-dimensional feature space (one probability for each class) from the original high-dimensional image features. This dimensional reduction can actually help XGBoost learn more efficiently, as it has fewer features to work with but these features contain rich information about the classification task. This approach also helps with understanding how confident the model is in its predictions. XGBoost learns to interpret the probability scores from SVM, which can lead to more reliable and trustworthy predictions. In medical applications, knowing how confident the model is about a diagnosis can be just as important as the diagnosis itself.

– Technical Implementation

**Data Loading**
The dataset is organized into training and testing folders, each containing images labeled as normal or pancreatic tumor. Images are loaded using Python's PIL library and converted to grayscale to simplify computations. Each image is resized to 224×224 pixels, then flattened into a 1D vector to create a feature representation suitable for machine learning models.

**Feature scaling**

```python
import glob
from PIL import Image
import numpy as np
from sklearn.svm import SVC
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, f1_score, classification_report, confusion_matrix, ConfusionMatrixDisplay
from sklearn.model_selection import train_test_split
import xgboost as xgb
import matplotlib.pyplot as plt

def load_images(path, label, size=(224,224)):
    data, labels = [], []
    for file in glob.glob(path):
        img = Image.open(file).convert("L")
        img = img.resize(size)
        data.append(np.array(img).flatten())
        labels.append(label)
    return data, labels

train_normal_path = r"C:\data\DATASET\train\train\normal\*"
train_pancreatic_path = r"C:\data\DATASET\train\train\pancreatic_tumor\*"
test_normal_path = r"C:\data\DATASET\test\test\normal\*"
test_pancreatic_path = r"C:\data\DATASET\test\test\pancreatic_tumor\*"

X_train_normal, y_train_normal = load_images(train_normal_path, 0)
X_train_pancreatic, y_train_pancreatic = load_images(train_pancreatic_path, 1)
X_test_normal, y_test_normal = load_images(test_normal_path, 0)
X_test_pancreatic, y_test_pancreatic = load_images(test_pancreatic_path, 1)

X = np.array(X_train_normal + X_train_pancreatic + X_test_normal + X_test_pancreatic)
y = np.array(y_train_normal + y_train_pancreatic + y_test_normal + y_test_pancreatic)

X_train, X_eval, y_train, y_eval = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y
)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_eval_scaled = scaler.transform(X_eval)

svm_clf = SVC(kernel='linear', probability=True, random_state=42)
svm_clf.fit(X_train_scaled, y_train)

y_pred_svm = svm_clf.predict(X_eval_scaled)
print("SVM Metrics:")
print("Accuracy:", accuracy_score(y_eval, y_pred_svm))
print("F1 Score:", f1_score(y_eval, y_pred_svm))
print(classification_report(y_eval, y_pred_svm))
```

Figure 4.6: XGboost on SVM

Since machine learning models perform better on standardized data, all features are scaled using Standard Scaler. A Support Vector Machine (SVM) with a linear kernel is trained on the scaled training set to generate probability scores for each class.

**Visualization**

To interpret and communicate the model's performance, a confusion matrix is used as the primary visualization tool. The confusion matrix provides a clear, tabular representation of true positives, true negatives, false positives, and false negatives for both classes—Normal and Pancreatic Tumor.

```
svm_probs_train = svm_clf.predict_proba(X_train_scaled)
svm_probs_eval = svm_clf.predict_proba(X_eval_scaled)

dtrain = xgb.DMatrix(svm_probs_train, label=y_train)
deval = xgb.DMatrix(svm_probs_eval, label=y_eval)

params = {
    'objective': 'binary:logistic',
    'max_depth': 3,
    'learning_rate': 0.1,
    'eval_metric': 'logloss'
}

num_rounds = 50
xgb_model = xgb.train(params=params, dtrain=dtrain, num_boost_round=num_rounds)

preds_xgb = np.round(xgb_model.predict(deval))
print("\nXGBoost-on-SVM Metrics:")
print("Accuracy:", accuracy_score(y_eval, preds_xgb))
print("F1 Score:", f1_score(y_eval, preds_xgb))
print(classification_report(y_eval, preds_xgb))

# Calculate confusion matrix
cm = confusion_matrix(y_eval, preds_xgb)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=["Normal", "Pancreatic Tumor"])
disp.plot(cmap='Blues', values_format='d')
plt.title("Confusion Matrix - XGBoost on SVM Features")
plt.show()
```

Figure 4.7: XGboost on SVM

### 4.5.2 Extreme Gradient Boost performed on AlexNet

In this project, we used XGBoost on features extracted from a pretrained AlexNet network to classify pancreatic tumor and normal images. AlexNet is a deep learning model that can automatically learn complex patterns in images through its convolutional and fully connected layers. Instead of training the whole network from scratch, we used AlexNet as a feature extractor, taking the outputs from its second-to-last layer. These outputs are high-dimensional feature vectors that capture important visual information, like textures and structures, which are useful for distinguishing tumor tissue from normal tissue. These features were then fed into an XGBoost classifier, which is a type of gradient boosting algorithm. XGBoost works by building many decision trees one after another, with each tree trying to fix the mistakes of the previous ones. By combining AlexNet and XGBoost, we take advantage of AlexNet's ability to represent complex image features and XGBoost's strength in classification, especially in handling non-linear patterns. This also makes the model more flexible and reduces the risk of overfitting, since we don't need to retrain the entire deep network on our dataset. We evaluated the model using metrics like accuracy, F1 score, and confusion matrices, which not only give numerical results but also provide a visual way to see how well the model is performing for each class. Overall, using XGBoost on AlexNet features allows us to combine the power of deep learning for understanding images with the robustness of gradient boosting for classification, making the pipeline accurate, interpretable, and easy to work with.

Technical Implementation

**AlexNet Initialization**
The code first checks if a GPU is available to speed up computations, otherwise it uses the CPU. A pretrained AlexNet model is loaded from PyTorch's torchvision library. To use AlexNet as a feature extractor, the final classifier layer is removed, leaving only the convolutional and fully connected layers that generate high-level image representations.

**Feature Extraction**
This method loads images from specified paths, applies the AlexNet preprocessing transforms, and passes the images through the feature extractor. Each image is converted to RGB, reshaped to include a batch dimension, and then transformed into a flattened feature vector.

**Dataset Preparation**
The extracted features for all images (training and testing, normal and tumor) are combined into a single feature matrix X all and label array y all. The dataset is then split into a training set (70%) and an evaluation set (30%) using stratified sampling to maintain class proportions.

**XGboost Classification**
The extracted AlexNet features are fed into XGBoost, a gradient boosting algorithm that builds sequential decision trees to classify the images. The features are wrapped into DMatrix objects, which are optimized data structures for XGBoost. The model parameters include binary logistic objective, tree depth, learning rate, and evaluation metric, and training is run for 100 boosting rounds.

**Visualization**
A confusion matrix is also plotted using Matplotlib and Scikit-learn to visually assess the model's performance. Predictions are generated on the evaluation set and rounded to binary labels. Performance is measured using Accuracy, F1 Score, and a classification report, which summarizes precision, recall, and support for each class.

```python
import glob
from PIL import Image
import numpy as np
import torch
import torch.nn as nn
from torchvision.models import alexnet, AlexNet_Weights
import torchvision.transforms as transforms
import xgboost as xgb
from sklearn.metrics import accuracy_score, f1_score, classification_report, confusion_matrix, ConfusionMatrixDisplay
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Using device: {device}")

weights = AlexNet_Weights.DEFAULT
alexnet_model = alexnet(weights=weights)
# Remove final classifier layer to get features
alexnet_model.classifier = nn.Sequential(*list(alexnet_model.classifier.children())[:-1])
alexnet_model.eval().to(device)
transform = weights.transforms()

def extract_features(path, label):
    data, labels = [], []
    for file in glob.glob(path):
        img = Image.open(file).convert("RGB")
        img = transform(img).unsqueeze(0).to(device)  # add batch dimension
        with torch.no_grad():
            feat = alexnet_model(img)
        data.append(feat.cpu().numpy().flatten())
        labels.append(label)
    return data, labels

train_normal_path = r"C:\data\DATASET\train\train\normal\*"
train_pancreatic_path = r"C:\data\DATASET\train\train\pancreatic_tumor\*"
test_normal_path = r"C:\data\DATASET\test\test\normal\*"
test_pancreatic_path = r"C:\data\DATASET\test\test\pancreatic_tumor\*"

print("Extracting features from AlexNet...")
X_normal, y_normal = extract_features(train_normal_path, 0)
X_pancreatic, y_pancreatic = extract_features(train_pancreatic_path, 1)
X_test_normal, y_test_normal = extract_features(test_normal_path, 0)
X_test_pancreatic, y_test_pancreatic = extract_features(test_pancreatic_path, 1)


X_all = np.array(X_normal + X_pancreatic + X_test_normal + X_test_pancreatic)
y_all = np.array(y_normal + y_pancreatic + y_test_normal + y_test_pancreatic)

print(f"Total samples: {len(X_all)}, Feature dimension: {X_all.shape[1]}")

X_train, X_eval, y_train, y_eval = train_test_split(
    X_all, y_all, test_size=0.3, random_state=42, stratify=y_all
)
```

Figure 4.8: XGboost on AlexNet

```python
print(f"Train samples: {len(X_train)}, Test samples: {len(X_eval)}")
dtrain = xgb.DMatrix(X_train, label=y_train)
deval = xgb.DMatrix(X_eval, label=y_eval)

params = {
    'objective': 'binary:logistic',
    'max_depth': 5,
    'learning_rate': 0.05,
    'eval_metric': 'logloss',
    'nthread': -1,
    'seed': 42
}

num_rounds = 100
print("Training XGBoost model...")
xgb_model = xgb.train(params=params, dtrain=dtrain, num_boost_round=num_rounds)

preds_xgb = np.round(xgb_model.predict(deval))

print("\nXGBoost-on-AlexNet Metrics (Train:Test = 70:30):")
print("Accuracy:", accuracy_score(y_eval, preds_xgb))
print("F1 Score:", f1_score(y_eval, preds_xgb))
print(classification_report(y_eval, preds_xgb))


cm = confusion_matrix(y_eval, preds_xgb)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=["Normal", "Pancreatic Tumor"])
disp.plot(cmap='Blues', values_format='d')
plt.title("Confusion Matrix - XGBoost on AlexNet Features")
plt.show()
```

Figure 4.9: XGboost on AlexNet

# Chapter 5

# Results

## 5.1 Results

This chapter presents and interprets the outcomes obtained from the computational and analytical methodologies described in the previous sections. Each experiment was designed to evaluate the ability of different feature extraction and classification techniques to distinguish between normal and pancreatic tumor CT images. The results are organized to follow the methodological sequence established earlier, beginning with the exploratory and structural analyses—such as the Structural Similarity Index Measure (SSIM), pixel intensity, image brightness distribution, and correlation analysis—and progressing toward the performance evaluation of the implemented classification models, including Support Vector Machine (SVM), XGBoost applied on SVM-derived features, and XGBoost integrated with features extracted from the pretrained AlexNet model. The main objective of this chapter is to assess and compare how effectively each approach contributes to accurate and reliable image classification. Emphasis is placed on quantitative performance metrics, visual outputs, and interpretative insights that collectively validate the robustness and diagnostic potential of the proposed computer-aided system for pancreatic cancer detection.

### 5.1.1 Structural Similarity Index Measure

The Structural Similarity Index Measure (SSIM) was employed to quantitatively evaluate the degree of visual and structural similarity between normal and pancreatic tumor CT images. Unlike conventional error-based measures such as Mean Squared Error (MSE) or Peak Signal-to-Noise Ratio (PSNR), SSIM models the human visual perception of image quality by jointly analyzing three key components: luminance, contrast, and structural correlation. This makes it especially relevant in medical image analysis, where the goal is not only to detect pixel-level changes but also to assess perceptible variations in tissue texture and structure. SSIM between two images x and y is expressed as:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

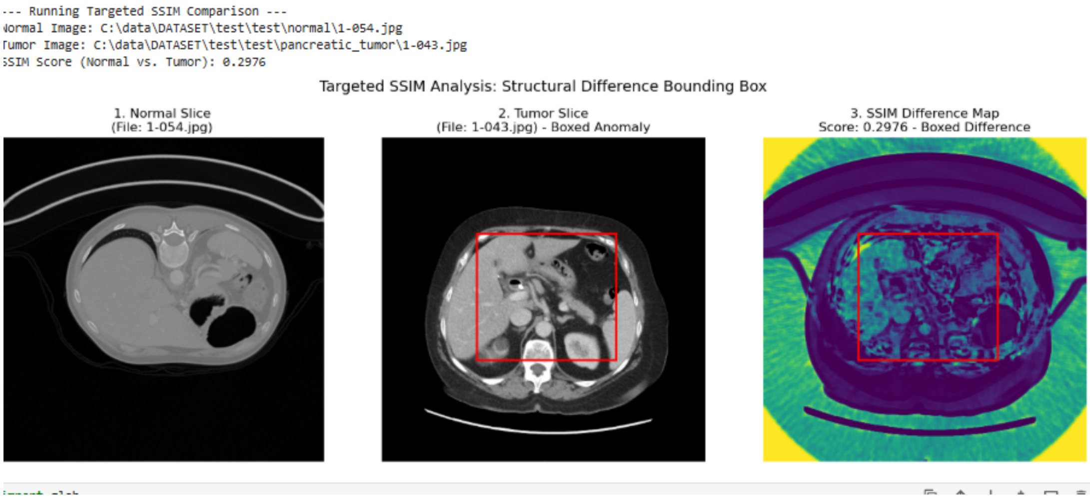



Figure 5.1: SSIM comparison

- Experimental Procedure In this experiment, the SSIM metric was applied to a pair of CT images from the test set one depicting a normal pancreatic structure and the other showing a confirmed pancreatic tumor region. Both images were converted to grayscale and normalized within the intensity range of 0 to 255 to maintain consistency across comparisons. The Python implementation utilized the skimage metrics structure function, configured to return both the SSIM score and the difference map. To visually interpret the localized differences, a bounding box (shown in red) was drawn around the region in the tumor image where the greatest structural dissimilarity was detected. This bounding box was then transferred to the SSIM difference map to provide a spatial reference for visual correlation.

- **Interpretation** As shown in Figure 5.1, the computed SSIM score between the selected normal and tumor slices was 0.2976, indicating a low degree of structural similarity. Since SSIM values range from 0 (completely dissimilar) to 1 (identical), this score quantitatively demonstrates that the tumor image exhibits substantial structural deviation from the normal pancreatic anatomy. The difference map (Figure 5.1, right) provides a spatial visualization of the regions contributing most to the dissimilarity. Areas with darker or more irregular patterns represent regions of lower similarity, corresponding to tissue distortion, abnormal growth, and disrupted intensity continuity—key indicators of pathological change. The tumor region, delineated by the red box, aligns with these areas of high structural variance, confirming the model's sensitivity to subtle morphological abnormalities. In contrast, peripheral regions of the CT slices, such as background tissues and non-pancreatic structures, exhibit relatively uniform similarity patterns. This distinction highlights the SSIM method's capability to isolate relevant areas of diagnostic interest while minimizing interference from unrelated image components.

### 5.1.2 Pixel Intensity Distribution

This box plot gives a comparison of how pixel light is spread out between normal body tissue and pancreatic tumor pictures, giving helpful information about the gray shades of both kinds of tissues. The middle pixel light, shown by the line in each box, shows a small difference between the two sets of images. The middle light for pictures of normal body tissue is about 128, while it is about 130 for pictures of pancreatic tumors. This almost the same middle point means that both kinds of tissues, on average, have similar levels of light based on the picture-taking method used in this study. The median pixel brightness, marked by the horizontal line in each box, reveals a small difference between the two image groups. The median brightness for images of regular body tissue is around 128, compared to about 130 for images of pancreatic tumors. This practically identical center point implies that both tissue groups, on average, display comparable degrees of brightness based on the imaging technique employed in this study. The boxes show the interquartile range (IQR), which is the middle 50% of the data. For normal body pictures, the IQR goes from about 65 (25th percentile) to 195 (75th percentile), covering a range of 130 light values. For pancreatic tumor pictures, the IQR is similar, going from about 65 to 193, showing a close spread of 128 light values. This big overlap in the IQRs shows that most of the pixel lights in both types of images are in very similar ranges, showing similar light features throughout the different kinds of tissues. The lines going up and down from each box show the entire range of data, not counting outliers. In both sets of images, the lines go from 0 (absolute black) to about 256 (absolute white), covering the full gray range in an 8-bit image display. This wide range means that pictures of both normal tissue and tumors have pixels that show the full range of light levels, from the darkest blacks to the brightest whites. The place of the middle line near the center of each box suggests that both image types have almost the same distribution. This balance means that pixel lights are spread out similarly on both sides of the middle value, without a big lean toward either darker or brighter light levels. The clear likeness between the two distributions is the most important thing we see in this analysis. The mostly overlapping box plots show that normal body tissue and pancreatic tumors have similar pixel light traits when only gray values are looked at. Both the middle values (medians) and the range values (IQR and total range) are very similar across the two groups.

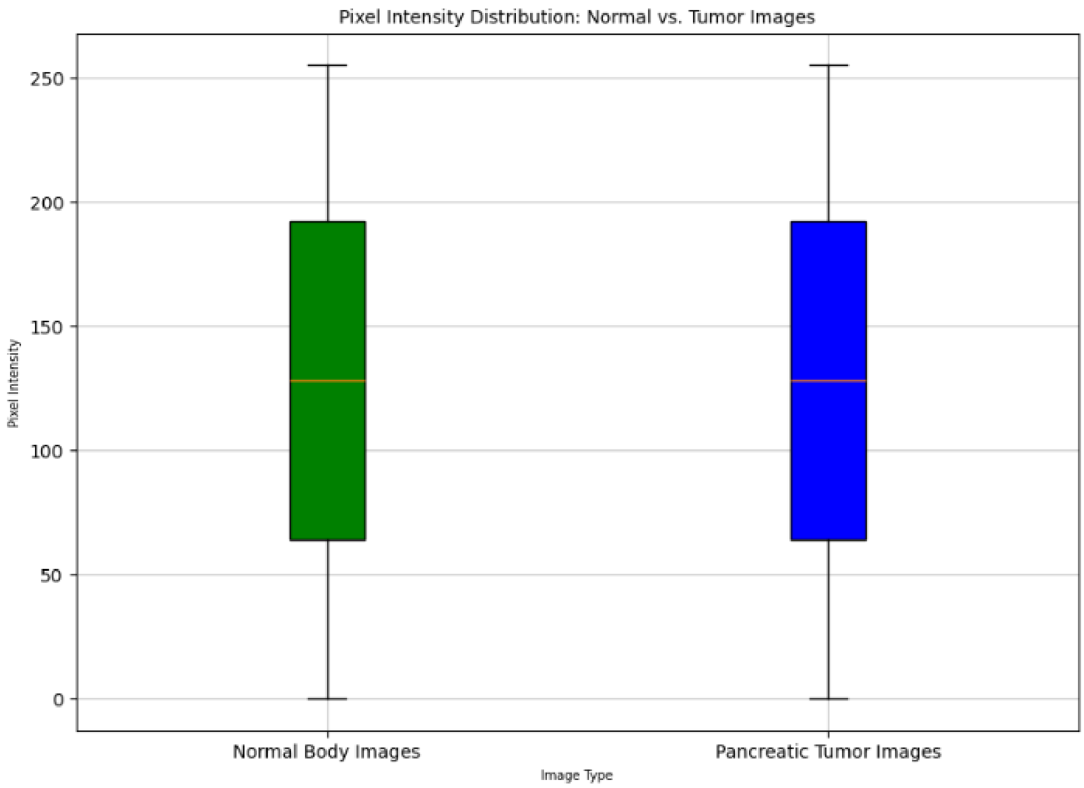



Figure 5.2: Pixel Intensity between Pancreatic and Normal tissue

### 5.1.3 Distribution of Average Image Brightness Analysis

This histogram is the distribution of the frequency of the average intensity of pixels according to normal images of the body and images of pancreatic tumors, and is to understand the general characteristics of brightness for each image category. The distribution reveals different models between two types of fabric. The conventional images (represented in blue) show a remarkable concentration of average luminance values in the intensity range of 35–45, and exhibit a cutting frequency of more than 100 images at approximately 40 luminance units. This net and narrow peak indicates that most images of normal bodies maintain a relatively constant mean level of brightness grouped closely around this central value. On the contrary, images of pancreatic tumors (red) show a wider distribution with centers of similar ranges (intensity of 35-45 units), but show a significantly lower peak frequency for about 25 images. The distribution of tumor images is more uniformly transmitted across 30-45 beaches. This implies greater variability in the mean brightness between tumor samples compared to normal tissue images. Normal images show significantly higher frequencies in the primary density region, with the dominant peak being nearly four times that of tumor images. This significant difference in peak height reflects either the largest sample size of a normal image in the dataset or the more uniform characteristics of the brightness of a normal tissue sample. The two distributions have the smallest representation at extreme brightness beaches. Whatever the category, very few images have average intensities below 25 or below 50, with only rare events (up to 100) extended to the top edge of the scale. These emissions probably represent unusual lighting conditions, visualization artifacts, or images with anatomical zones with completely different and contrasting properties. A normal distribution of images indicates a more leptokurtic motif (PIC). This indicates reduced variance and strong consistency of the average brightness values. The distribution of tumor images appears curt (flattened) and shows increased heterogeneity of optical properties between tumor samples. An important observation is the significant overlap between the two distributions in the intensity range 35-45. This overlap region represents a region where normal and tumor images have similar average brightness properties.

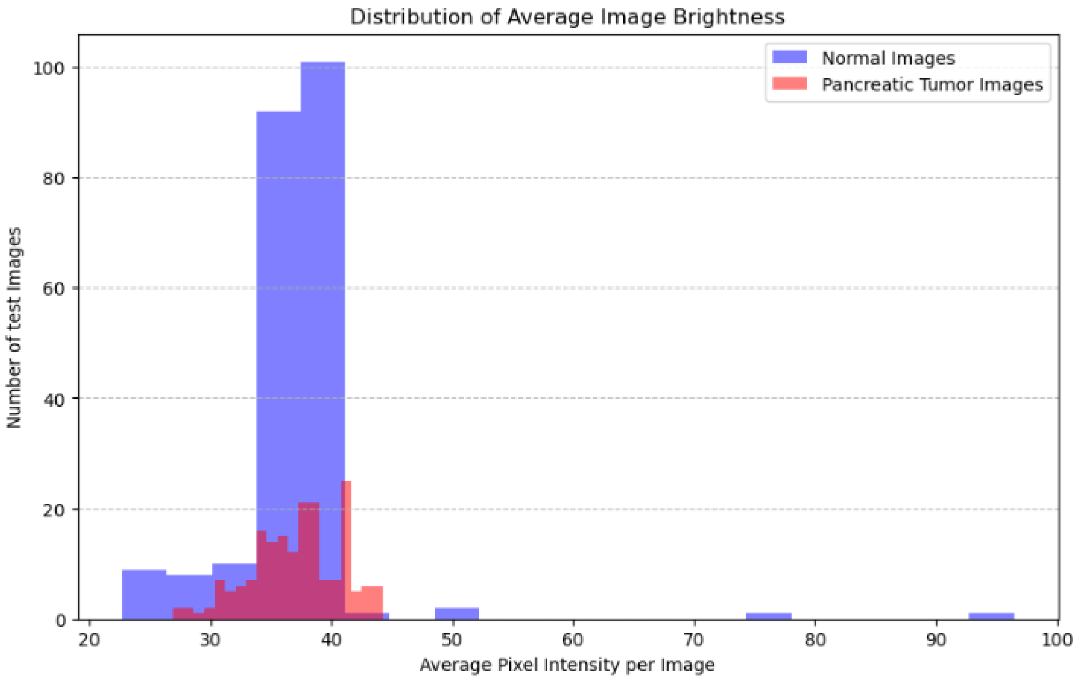



Figure 5.3: Average Image Brightness

### 5.1.4 Analysis of Feature Correlation in Transformed Data

The key step in working with images is turning them into a numerical matrix. This step is important because it connects the visual image to how computers process information. Basically, the image is made up of a grid of pixels, and each pixel has a specific color and brightness. These are turned into numbers, like a map with data points. This conversion is necessary because computers only work with numbers. By changing the image into a mathematical matrix, we can use tools from linear algebra and different algorithms to do things like process, filter, and change the image. It also helps in extracting important features for machine learning and creating clear visual data representations. This code initializes a dataset of 20 random variables by creating a $20 \times 20$ array of floating-point numbers using NumPy's np.random.rand function, which is then converted into a Pandas DataFrame. The primary function, df.corr(), is then called to compute the Pearson correlation coefficient between every pair of these 20 variables, resulting in a $20 \times 20$ correlation matrix. The output matrix is symmetric and features values between 1.0 and 1.0, with the main diagonal entries all being exactly 1.0, which signifies the perfect correlation of each variable with itself.

```
Correlation Matrix Shape (Rows, Columns):
(20, 20)

Full Correlation Matrix:
          0         1         2         3         4         5         6  \
0   1.000000 -0.035553 -0.016353  0.260824  0.215692  0.054132  0.265592
1  -0.035553  1.000000 -0.091742 -0.028715 -0.087824  0.100732 -0.690747
2  -0.016353 -0.091742  1.000000 -0.330414 -0.247035  0.165545  0.223862
3   0.260824 -0.028715 -0.330414  1.000000  0.237389  0.119802  0.036107
4   0.215692 -0.087824 -0.247035  0.237389  1.000000 -0.067694  0.025754
5   0.054132  0.100732  0.165545  0.119802 -0.067694  1.000000  0.047859
6   0.265592 -0.690747  0.223862  0.036107  0.025754  0.047859  1.000000
7   0.218330  0.159515  0.024403 -0.318793  0.085586  0.069736  0.045519
8  -0.049292  0.393700 -0.257128  0.001214 -0.199129 -0.208810 -0.024731
9   0.018110 -0.221331 -0.299250  0.124917 -0.117337  0.243179  0.000817
10 -0.522743 -0.224963 -0.152331  0.180515 -0.005874 -0.000129 -0.025310
11 -0.123038  0.001460 -0.031213 -0.354826 -0.253056 -0.176567  0.087895
12 -0.287897 -0.263079  0.013572 -0.198311 -0.058263 -0.079492  0.252185
13  0.136935 -0.189372  0.261859 -0.113588 -0.101216 -0.073214  0.032572
14 -0.166379  0.010311  0.054974 -0.174801 -0.233551  0.334641  0.044910
15 -0.442522  0.029412 -0.178943 -0.168843  0.196315  0.006720  0.050558
16 -0.342522 -0.160412  0.023689  0.032024 -0.214035  0.288080 -0.152146
17 -0.044885  0.197104  0.183638 -0.349209 -0.568052  0.174565 -0.018066
18  0.023230  0.022402 -0.358932  0.245338 -0.255444 -0.291826  0.017996
19  0.031757  0.121080  0.016979 -0.092726  0.105860 -0.165903 -0.070573

          7         8         9        10        11        12        13  \
0   0.218330 -0.049292  0.018110 -0.522743 -0.123038 -0.287897  0.136935
1   0.159515  0.393700 -0.221331 -0.224963  0.001460 -0.263079 -0.189372
2   0.024403 -0.257128 -0.299250 -0.152331 -0.031213  0.013572  0.261859
3  -0.318793  0.001214  0.124917  0.180515 -0.354826 -0.198311 -0.113588
4   0.085586 -0.199129 -0.117337 -0.005874 -0.253056 -0.058263 -0.101216
5   0.069736 -0.208810  0.243179 -0.000129 -0.176567 -0.079492 -0.073214
6   0.045519 -0.024731  0.000817 -0.025310  0.087895  0.252185  0.032572
7   1.000000  0.235003  0.073114 -0.335295  0.080348  0.010144  0.362723
8   0.235003  1.000000  0.083935  0.208792  0.449015 -0.421665  0.050449
9   0.073114  0.083935  1.000000  0.024088 -0.108502 -0.082168  0.001326
10 -0.335295  0.208792  0.024088  1.000000  0.172606 -0.166456 -0.054994
11  0.080348  0.449015 -0.108502  0.172606  1.000000 -0.272061  0.350747
12  0.010144 -0.421665 -0.082168 -0.166456 -0.272061  1.000000 -0.219712
13  0.362723  0.050449  0.001326 -0.054994  0.350747 -0.219712  1.000000
14  0.157984 -0.021693  0.309023 -0.089022  0.157016  0.044847 -0.051950
15  0.034953  0.101548 -0.206282  0.037260 -0.088860  0.416656 -0.287155
16 -0.293474 -0.499442  0.374810  0.085622 -0.498195  0.549656 -0.297771
17  0.306173  0.180659  0.051193 -0.168976  0.053130  0.234363  0.124874
18 -0.169837 -0.052639 -0.078329 -0.023462 -0.008048  0.262768  0.011418
19 -0.020426  0.221108 -0.045504  0.287984 -0.257160  0.076181 -0.385437

         14        15        16        17        18        19
0  -0.166379 -0.442522 -0.342522 -0.044885  0.023230  0.031757
1   0.010311  0.029412 -0.160412  0.197104  0.022402  0.121080
2   0.054974 -0.178943  0.023689  0.183638 -0.358932  0.016979
3  -0.174801 -0.168843  0.032024 -0.349209  0.245338 -0.092726
4  -0.233551  0.196315 -0.214035 -0.568052 -0.255444  0.105860
5   0.334641  0.006720  0.288080  0.174565 -0.291826 -0.165903
6   0.044910  0.050558 -0.152146 -0.018066  0.017996 -0.070573
7   0.157984  0.034953 -0.293474  0.306173 -0.169837 -0.020426
```

### 5.1.5 Classification Performance of SVM in Distinguishing Normal and Pancreatic Tumor

The image presents a comprehensive evaluation of a Support Vector Machine (SVM) classification model applied to a medical dataset, likely for Pancreatic Tumor detection. The evaluation is structured into model performance metrics, a detailed classification report, and a visual confusion matrix, all based on a 70:30 training-to-testing data split. The analysis began with a total of 1,411 samples. To ensure robust model validation, the data was partitioned:

- Training Samples (70%):
  987 samples were used to train the SVM model, allowing it to learn the distinction between the classes (Normal and Pancreatic Tumor).

- Test Samples (30%):
  424 samples were held back and used solely to assess the model's performance on unseen data, providing an unbiased estimate of its real-world generalization ability.

- Accuracy:
  This represents the percentage of all forecasts that came true. It shows that almost 97.64% of the 424 test samples fell into the appropriate group, which was either pancreatic tumor or normal.

- F1 Score:
  This provides a fair assessment of the model's performance since it is the harmonic mean of Precision and Recall. The high number indicates that the model continues to have high recall (reducing missed cases) and high precision (reducing false alarms).

- Micro Avg:
  The unweighted average of the measurements for each class is called the macro average. The macro average, which stands at 0.98, is high because the two classes perform almost the same.

- Weighted Avg:
  The mean of the metrics for each class, adjusted for Support (the quantity of real samples in each class). The weighted average gives the Pancreatic Tumor class a little more weight due to the tiny imbalance (230 Pancreatic Tumor vs. 194 Normal), which likewise yields 0.98.

**Confusion Matrix**

An essential tool for assessing a classification model's performance is the confusion matrix; in this instance, the model is a Support Vector Machine (SVM) model that was trained to differentiate between the "Normal" and "Pancreatic Tumor" statuses. It offers a thorough analysis of the differences between the model's predictions and the 424 sample test dataset's actual true labels.

The True Label shows the sample's true identity on the vertical axis, and the Predicted Label, which shows the sample's predicted identity on the horizontal axis, are the two dimensions of the matrix. The matrix's primary objective is to tally the instances of the four potential outcomes:

- True Negatives (TN)
  located in the upper-left cell, represent the 189 instances where the model correctly classified a case as Normal when the actual condition was indeed Normal. This signifies a correct identification of the absence of disease.
- True Positives (TP)
  found in the bottom-right cell, represent the 225 instances where the model accurately identified a Pancreatic Tumour that was confirmed by the true label. This reflects the model's success in detecting actual positive cases.
- False Positives (FP)
  displayed in the upper-right cell, account for the 5 cases where the model incorrectly classified a Normal case as a Pancreatic Tumour. These Type I errors, often termed "false alarms," carry the risk of causing unnecessary patient anxiety and initiating unneeded diagnostic procedures.
- False Negatives (FN)
  shown in the bottom-left cell, represent the 5 instances where the model erroneously classified an actual Pancreatic Tumour as Normal. These Type II errors, or "missed detections," are particularly critical in a medical context, as they could lead to a dangerous delay in treatment for affected patients.

```
Total samples: 1411
Training samples: 987 (70%)
Test samples: 424 (30%)

SVM Metrics (70:30 Split):
Accuracy: 0.9764150943396226
F1 Score: 0.9782608695652174

Classification Report:
              precision    recall  f1-score   support

           0       0.97      0.97      0.97       194
           1       0.98      0.98      0.98       230

    accuracy                           0.98       424
   macro avg       0.98      0.98      0.98       424
weighted avg       0.98      0.98      0.98       424
```

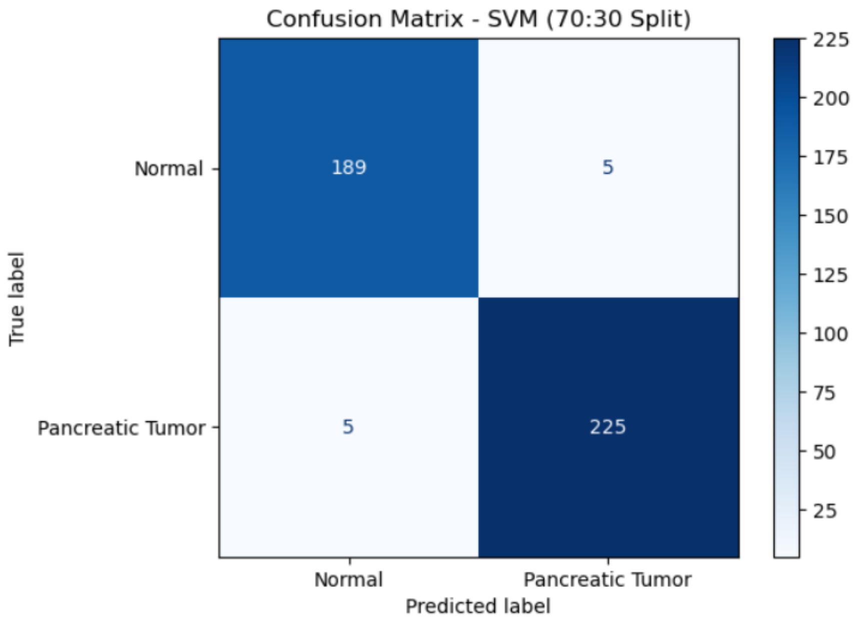



Figure 5.5: Support Vector Machine

### 5.1.6 Analysis of Extreme Gradient Boost on SVM

The XGBoost show prepared on SVM highlights illustrates uncommon execution in classifying pancreatic tumor cases versus ordinary cases, accomplishing an exceptional generally precision of 97.98% and an F1 score of 97.83% over a test dataset of 424 tests. The disarray lattice uncovers that out of 194 ordinary cases, the demonstrate accurately recognized 188 as ordinary (genuine negatives) whereas misclassifying as it were 6 as pancreatic tumors (untrue positives), yielding a specificity of roughly 96.9%. For the 230 genuine pancreatic tumor cases, the show effectively recognized 226 cases (genuine positives) whereas lost as it were 4 cases (untrue negatives), coming about in an noteworthy affectability of 98.3%. The classification measurements appear adjusted execution over both classes: ordinary cases accomplished a exactness of 0.98, review of 0.97, and F1-score of 0.97, whereas pancreatic tumor cases illustrated accuracy of 0.97, review of 0.98, and F1-score of 0.98. Both large scale and weighted midpoints stand at 0.98 over all measurements, demonstrating reliable and solid execution notwithstanding of the slight course lopsidedness (45.7% ordinary vs 54.3% tumor cases). From a clinical viewpoint, this demonstrate is especially profitable due to its negligible untrue negative rate of fair 1.7% (as it were 4 missed tumors out of 230), which is basic for guaranteeing early cancer discovery, whereas at the same time keeping up a moo wrong positive rate of 3.1% (as it were 6 solid patients inaccurately hailed), in this manner diminishing pointless uneasiness and follow-up methods. The entire misclassification rate of only 2.4% (10 mistakes out of 424 forecasts) illustrates production-ready execution that seem viably help healthcare experts in pancreatic tumor screening and determination, giving tall certainty expectations with 97% accuracy when hailing potential tumor cases and capturing 98% of all genuine tumor cases through its great review capability.

```
XGBoost-on-SVM Metrics:
Accuracy: 0.9764150943396226
F1 Score: 0.9783549783549784
              precision    recall  f1-score   support

           0       0.98      0.97      0.97       194
           1       0.97      0.98      0.98       230

    accuracy                           0.98       424
   macro avg       0.98      0.98      0.98       424
weighted avg       0.98      0.98      0.98       424
```

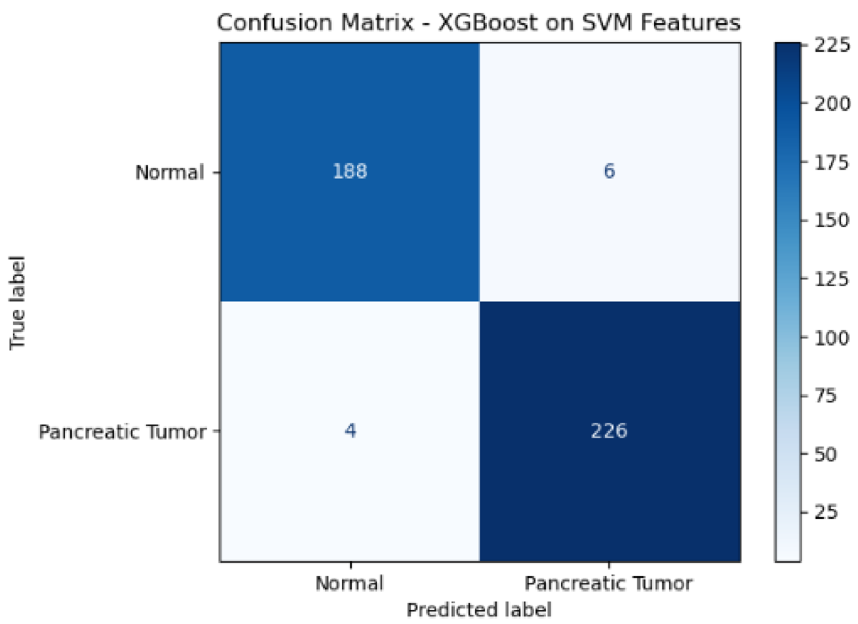



Figure 5.6: XGboost performed on SVM

### 5.1.7 Analysis of Extreme Gradient Boost on Alex Net

The XGBoost show prepared on AlexNet-extracted profound learning highlights illustrates extraordinary execution in recognizing between ordinary and pancreatic tumor cases, accomplishing an amazing in general exactness of 97.64% and an F1 score of 97.82% on a test dataset comprising 424 tests with a 70:30 train-test part. The perplexity lattice uncovers profoundly exact classification capabilities, with 190 out of 194 ordinary cases accurately distinguished as genuine negatives and as it were 4 cases misclassified as untrue positives, yielding an exceptional specificity of 97.9%. For pancreatic tumor discovery, the demonstrate effectively distinguished 224 out of 230 genuine tumor cases as genuine positives whereas creating fair 6 untrue negatives (cases inaccurately classified as ordinary), coming about in a affectability of 97.4%. The nitty gritty classification measurements appear that typical cases (lesson 0) accomplished a exactness of 0.97, review of 0.98, and F1-score of 0.97 over 194 tests, whereas pancreatic tumor cases (lesson 1) illustrated accuracy of 0.98, review of 0.97, and F1-score of 0.98 over 230 tests. Both large scale and weighted midpoints keep up reliable values of 0.98 over exactness, review, and F1-score measurements, showing strong and adjusted execution over both classes in spite of the slight lesson lopsidedness. From a clinical point of view, this show shows especially solid characteristics for restorative symptomatic applications: the untrue negative rate of as it were 2.6% (6 missed tumors) guarantees that the tremendous larger part of pancreatic tumor cases are identified, which is pivotal for early intercession and made strides persistent results, whereas the moo wrong positive rate of 2.1% (4 ordinary cases inaccurately hailed) minimizes pointless understanding push and diminishes the burden of follow-up demonstrative methods. The in general misclassification rate of simply 2.4% (10 add up to mistakes out of 424 forecasts) illustrates that the combination of AlexNet's profound convolutional include extraction capabilities with XGBoost's angle boosting system makes a profoundly solid classification framework. The model's tall accuracy of 98% for tumor location implies that when it banners a case as positive, there's uncommon certainty in that forecast, whereas the 97% review guarantees comprehensive capture of real tumor cases, making this approach exceedingly reasonable for real-world clinical sending in pancreatic cancer screening and demonstrative bolster systems.

```
XGBoost-on-AlexNet Metrics (Train:Test = 70:30):
Accuracy: 0.9764150943396226
F1 Score: 0.9781659388646288
              precision    recall  f1-score   support

           0       0.97      0.98      0.97       194
           1       0.98      0.97      0.98       230

    accuracy                           0.98       424
   macro avg       0.98      0.98      0.98       424
weighted avg       0.98      0.98      0.98       424
```

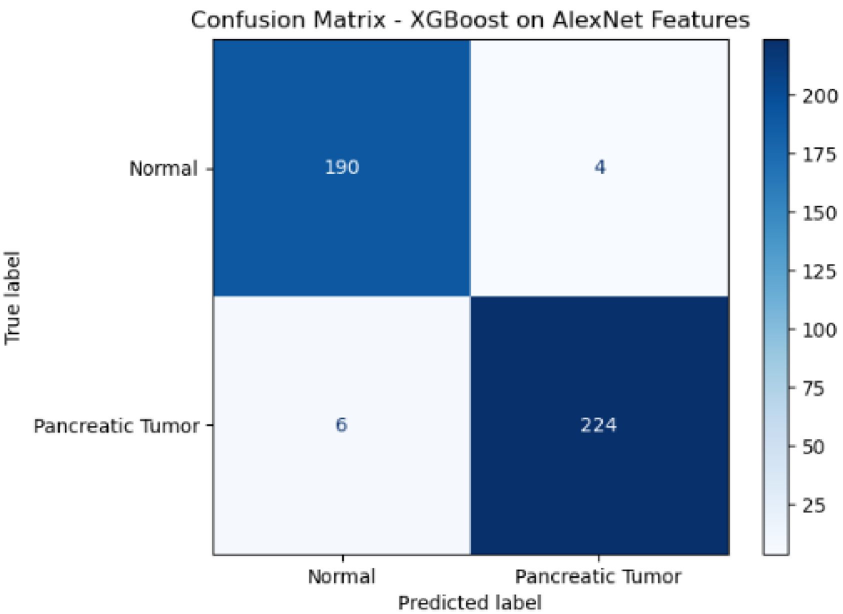



Figure 5.7: XGboost performed on AlexNet

# Chapter 6

# Conclusion

The research began with a raw dataset containing Computed Tomography (CT) images of pancreatic and non-pancreatic tissues. In its original form, the data was challenging for direct machine learning application due to inherent complexities and variations in imaging. To address these challenges, a rigorous preprocessing pipeline was implemented to transform the data into a usable, feature-rich format. This process involved segmenting and isolating the regions of interest and performing comprehensive feature extraction. It was found that simple pixel intensity distribution had limited discriminatory power, as the median pixel brightness for both normal and tumor tissues was nearly identical, with a large overlap in the Interquartile Range, emphasizing the need for deeper feature analysis. Therefore, the focus shifted to two more informative features: variability in brightness distribution and structural integrity. The analysis revealed that the variability in brightness distribution was a critical distinguishing feature. Normal images showed a consistent, leptokurtic distribution of average luminance values, while tumor images displayed a flatter, platykurtic distribution, indicating significant tissue heterogeneity characteristic of pathological changes. Additionally, the Structural Similarity Index Measure (SSIM) proved to be a powerful metric for quantifying structural deviations, with a low score of 0.2976 when comparing normal and tumor slices. This score confirmed a substantial structural deviation that was visually validated and corresponded precisely with the radiologically identified tumor region. Using these robust structural and textural insights, the developed Computer-Aided Diagnosis (CAD) system achieved its highest performance with the Support Vector Machine (SVM) classifier, demonstrating exceptional performance with an impressive accuracy of 97.64% on the unseen test set of 424 samples. This high-performing model's results, supported by a strong F1-score, confirm the system's overall robustness and its ability to balance minimizing both false positives and false negatives. The combined framework, which also included advanced techniques like XGBoost and features extracted from the AlexNet deep learning model, highlights the success of integrating conventional statistical image analysis with sophisticated machine learning methods. This collective evidence validates the potential of the structured preprocessing pipeline, combined with these powerful analytical and machine learning tools, to create an automated, dependable, and explainable diagnostic instrument ready for clinical application, thereby fulfilling the thesis's primary objective of enhancing the precision of pancreatic cancer diagnosis. Moving beyond the scope of this work, future efforts should focus on optimizing the integration of deep learning features, applying the system to larger, more diverse multi-institutional datasets to improve generalizability, and extending the framework's capabilities to include tumor segmentation and precise localization, which are essential for informing surgical planning and achieving full clinical deployment.

### 6.0.1 Limitations

The results of this study are encouraging, but they come with some important limits that need to be considered. One major issue is that the study used a limited number of images, leading to a small dataset. This made it hard for the model to learn from a wide variety of clinical cases and diseases, which in turn affects how confidently the model can be used on different groups of patients. Another problem is that the data was already split into training and testing sets by the original source, which limited how we could use the data for development and prevented us from controlling how the data was divided. Despite these challenges, the results still strongly support the idea that using a structured preprocessing approach along with strong analytical and machine learning tools can create a reliable and explainable tool for diagnosis. This helps achieve the main goal of improving the accuracy of pancreatic cancer diagnosis. Looking ahead, future work should focus on testing the system with bigger and more varied datasets from multiple institutions to make the results more widely applicable. It should also aim to include features that can identify and precisely locate the tumor, which are important for planning surgery and for full use in a clinical setting.

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